68

Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

Alan W. Shindel, and Tom F. Lue

Keywords

erectile dysfunction; erection; men; penis; sexual dysfunction

Questions

1. The thinnest portion of the tunica albuginea is the:
    a. lateral surface.
    b. ventral surface.
    c. crura.
    d. glans.
    e. dorsal surface.
2. The tunica albuginea of the corpora cavernosa consists of:
    a. concentric coats of nerves, arteries, and veins.
    b. an outer layer of circular fibers.
    c. type II collagen.
    d. valves to coapt emissary veins.
    e. a middle layer of obliquely oriented fibers.
3. Penile circulation is derived from both internal pudendal and accessory arteries in approximately what percentage of men?
    a. 5%

b. 30%
c. 50%
d. 70%
e. 90%

4. During the rigid phase of erection, all of the following are typically present except:
    a. arterial vasodilation.
    b. corporal sinusoid relaxation.
    c. corporal pressure increase.
    d. subtunical venous compression reducing venous outflow.
    e. relaxation of the ischiocavernosus muscles.
5. All of the following neurotransmitters are known to be highly relevant in male sexual function except:
    a. dopamine.
    b. oxytocin.
    c. vasoactive intestinal polypeptide (VIP).
    d. 5HT (serotonin).
    e. nitric oxide (NO).
6. All of the following statements are true except:
    a. NO stimulates the production of cyclic guanosine monophosphate (cGMP).
    b. NO released by endothelial nitric oxide synthase (eNOS) contained in the terminals of the cavernous nerve initiates the erection process, whereas nitric oxide released from neuronal nitric oxide synthase (nNOS) in the endothelium helps maintain erection.
    c. cGMP activates protein kinase G, which in turn opens potassium channels and closes calcium channels.
    d. low cytosolic calcium favors smooth muscle relaxation.
    e. smooth muscle regains its tone when cGMP is degraded by phosphodiesterase.
7. According to Endocrine Society guidelines, testosterone therapy is indicated when:
    a. total testosterone levels are less than 300 ng/dL.
    b. the patient has low libido.
    c. the patient requests it.

d. a dual-energy X-ray absorptiometry (DEXA) scan shows decreased bone mineral density.
e. none of the above.

8. Which of the following antihypertensives is thought to potentially exert a beneficial effect on erectile function?
   a. Angiotensin-converting enzyme inhibitors
   b. Thiazide diuretics
   c. β blockers
   d. Calcium channel blockers
   e. None of the above
9. Erectile dysfunction (ED) in end-stage renal disease (ESRD) has been commonly associated with all of the following except:
   a. reduced bioavailability of NO
   b. autonomic neuropathy
   c. low serum prolactin
   d. endothelial dysfunction
   e. arterial calcification
10. Sonic hedgehog (SHH) protein has been shown to:
   a. inhibit normal penile development.
   b. reduce apoptosis in animal models of cavernous nerve injury.
   c. reduce expression of vascular endothelial growth factor (VEGF) and brain-derived neurotrophic factor (BDNF).
   d. enhance erectile responses in men after radical prostatectomy.
   e. increase penile tissues after cavernous nerve injury.
11. Rho kinase:
   a. phosphorylates the regulatory subunit of myosin light chain phosphatase.
   b. phosphorylates the regulatory subunit of myosin light chain kinase.
   c. inactivates RhoA.
   d. reduces smooth muscle contraction.
   e. activates soluble guanylate cyclase.
12. All of the following are **false** statements about nitric oxide synthase (NOS) except:
   a. synthesis of NO is catalyzed by NOS, which converts L-citrulline and oxygen to L-arginine and NO.

b. upregulation of nNOS expression has been found in the corpus cavernosum of aging and diabetic rats.
c. iNOS is a principle contributor to NO production in physiologic penile erection.
d. nNOS localizes to tyrosine hydroxylase containing nerve fibers.
e. NOS exists as three isoforms in mammalian cells.

13. All the following statements are true except:
    a. C-type natriuretic peptide (CNP) is a potent natriuretic peptide; it relaxes isolated cavernous smooth muscle by binding to natriuretic peptide receptor-B (NPR-B)
    b. protein kinase G-I (PKGI) may induce relaxation via activation of the plasma membrane Ca^{2+}-ATPase pump, inhibition of inositol trisphosphate (IP_3) generation, inhibition of Rho-kinase, stimulation of myosin light chain phosphatase (MLCP), and phosphorylation of heat shock proteins
    c. reduced penile adenosine levels are associated with prolonged penile erection in animal models
    d. calcitonin gene–related peptide is a potent vasodilator released from perivascular nerve fibers
    e. the erectogenic effects of prostaglandin E_1 (PGE_1) as a pharmaceutic agent have been extensively documented
14. Relative to men without ED, psychogenic ED may be related to:
    a. excessive sympathetic tone.
    b. decreased activity of the “salience network.”
    c. increased activity of the “default mode network.”
    d. two of the above.
    e. all of the above.
15. The temporal and parietal lobes
    a. are activated during sexual arousal
    b. govern introspection and self-reflection
    c. trigger orgasmic response
    d. are essential for cognitive evaluation of erotic stimuli
    e. none of the above

16. All of the following statements about arteriogenic ED are **false** except:
    a. carbon monoxide inhibits erectile responses by inhibiting heme oxygenase
    b. protein kinase G inhibits myosin light chain phosphatase to induce penile erection
    c. hyperpolarization of smooth muscle cells is associated with penile erection
    d. prostaglandin PGI_2 induces penile erection
    e. hydrogen sulfide induces penile erection and is produced by breakdown of arginine
17. Cavernous nerve injury is associated with
    a. depletion of M_1 macrophages
    b. upregulation of SHH
    c. downregulation of RhoA/Rho-kinase
    d. decreased oxidative stress
    e. none of the above

Answers

1. **b. Ventral surface.** Histological studies have indicated that the outer longitudinal layer of the tunica is absent between the 5 and 7 o'clock position on the penis. For this reason, erosion of penile prosthetics are more common through the ventral tunica toward the urethra.
2. **e. A middle layer of obliquely oriented fibers.** The tunica albuginea is a tough fibrous sheath containing type I and III collagen fibers (no type II). The outer layer of the tunica consists of longitudinally oriented fibers and the inner layer consists of circular fibers. A middle layer of obliquely oriented fibers connects the outer and inner layers. Penile erection does not require presence of valves to restrict blood flow; the emissary veins are compressed between the erectile tissue and the tough fibrous sheath of the tunica.

3. **d. 70%.** The majority of men have blood supply from both the internal pudendal and the accessory pudendal arteries. Vascular studies indicate that about 15% of men have corporal blood supply derived exclusively from either the internal pudendal or the accessory pudendal arteries. These men may be more vulnerable to vascular injury than men with redundant circulation from both arteries.
4. **e. Relaxation of the ischiocavernosus muscles.** The rigid erection phase corresponds to maximal sexual arousal and penile turgidity. Arterial vasodilation, sinusoid relaxation, increased corporal pressure, and venous occlusion are components of the full erection phase that continue into the rigid erection phase. Maximal engorgement of the entire penis requires contraction of the ischiocavernosus muscles to force blood into the distal corpora, glans, and corpus spongiosum.
5. **c. Vasoactive intestinal polypeptide (VIP).** Nitric oxide is critical for initiation of vasodilation. Dopamine, oxytocin, and serotonin play important roles regulating sexual responses (libido, erection, ejaculation) in the central nervous system. VIP containing nerves and peptides have been identified in the penis of experimental animals, but VIP administration does not produce substantive changes in penile erection and does not appear to be essential for erection response.
6. **b. NO released by endothelial nitric oxide synthase (eNOS) contained in the terminals of the cavernous nerve initiates the erection process, whereas nitric oxide released from neuronal nitric oxide synthase (nNOS) in the endothelium helps maintain erection.** NO activates guanylate cyclase, which in turn produces cGMP, which activates protein kinase G. Degradation of cGMP leads to reversal of smooth muscle tone. eNOS is activated in part due to shear stress from vasodilation and helps to maintain erection; the erection must be initiated by another process, typically release of nitric oxide from cavernous nerves and nNOS.
7. **e. None of the above.** It is recommended that testosterone therapy only be administered in the setting of biochemically low levels (less than approximately 300 ng/dL) AND symptoms potentially

attributable to low testosterone (e.g., low libido, fatigue, loss of muscle mass, loss of bone density, depressed mood, weight gain). These symptoms are nonspecific and therefore do not in and of themselves merit T therapy but may improve with testosterone supplementation in the context of biochemically low levels.

8. **a. Angiotensin-converting enzyme inhibitors.** Thiazides and β blockers have been linked to ED. Angiotensin-converting enzyme (ACE) inhibitors do not have known effects that would interfere with erection, and there are data to suggest that there may be beneficial effects from ACE inhibition.
9. **c. Low serum prolactin.** Uremia and vascular changes are common in men with ESRD. There may be hormonal disruptions that contribute to ED in ESRD; however, it is typically elevated prolactin that plays a role in ED.
10. **b. Reduce apoptosis in animal models of cavernous nerve injury.** SHH is a trophic peptide that is essential in nervous system and musculoskeletal development. SHH enhances VEGF and BDNF expression; expression of SHH declines after cavernous nerve injury. While SHH has been associated with improvements in erectile function in animal models, it has not to date had proven efficacy in managing ED in human men who have undergone radical prostatectomy.
11. **a. Phosphorylates the regulatory subunit of myosin light chain phosphatase.** Rho kinase opposes penile erection by phosphorylating the regulatory subunit of MLCP, inhibiting it and maintaining phosphorylation of the myosin light chain. This has the effect of stimulating smooth muscle contraction and vasoconstriction, opposing penile erection. Rho kinase does not directly act on guanylate cyclase. Dissociation of RhoA from the Rho-guanine dissociation inhibitor leads to activation of Rho kinase, not vice versa.
12. **e. NOS exists as three isoforms in mammalian cells.** Answer a is close but reversed, as it is conversion of arginine and oxygen to citrulline and NO that is facilitated by NOS. Upregulation of nNOS is associated with superior erectile function, which is counter to what is observed commonly in diabetic and aged animals. nNOS

and eNOS are physiologically important whereas iNOS is activated in the setting of tissue stress. Tyrosine hydroxylase is an enzyme involved in production of catecholamines, which tend to favor sympathetic tone and oppose erectile processes. NOS does not co-localize with tyrosine hydroxylase in cavernous tissues and nerves.

13. **c. Reduced penile adenosine levels are associated with prolonged penile erection in animal models.** Adenosine can have either relaxant or constrictive effects on vascular tissue depending on the relative abundance of the various adenosine receptors in cells. Adenosine deaminase deficient mice and sickle cell transgenic mice experienced prolonged penile erections and are utilized as rodent models of the human condition of priapism. Accumulation of adenosine in these rodents is thought to be the mediator of these prolonged penile erections.
14. **d. Two of the above.** Sympathetic tone is associated with physiological and psychological stress and may contribute to impaired erectile response in psychogenic ED. The default mode network is the network of brain structures and connections thought to be involved with a state of "wakeful rest" and has been reported to exhibit decreased activity in men with ED thought to be psychogenic in origin. The salience network is involved in integrating and processing external stimuli and has been shown to be less active in men with psychogenic ED. Lower-level activity of this network may hypothetically compromise self-recognition of autonomic and sexual responses and thus inhibit sexual response.
15. **b. Govern introspection and self-reflection.** The temporal and parietal lobes are involved in processes of introspection and self-reflection. It is thought that decreased activity of these brain regions plays a role in disinhibition, which may facilitate sexual responses. These regions are not thought to play a role in orgasmic response, which appears to primarily involve the mesodiencephalic transition zone (including the ventral tegmental area), the midbrain lateral central tegmental field, the zona incerta, the subparafascicular nucleus, and the thalamic nuclei.
16. **c. Hyperpolarization of smooth muscle cells is associated with penile erection.** Hyperpolarization is associated with reduction in

intracellular calcium and subsequent inhibition of smooth muscle contraction; this has the net effect of facilitating penile erection. Carbon monoxide is produced by action of heme oxygenase and appears to facilitate erection response, possibly by binding to guanylate cyclase. Protein kinase G has the downstream effect of activating MLCP, which in turn works to deactivate the myosin light chain and oppose smooth muscle contraction. Prostaglandin E1 is associated with penile erection, but the PGI_2 isoform induces vasoconstriction, opposing erection response. Hydrogen sulfide induces penile erection but is produced by breakdown of L-cysteine.

17. **e. None of the above.** Cavernous nerve injury is associated with worsening of inflammation, tissue damage, and vascular response. All of the actions noted above would tend to facilitate penile erection response and/or preservation of penile tissues. M_1 macrophages are neurotoxic and are associated with tissue injury and remodeling. SHH is associated with nerve and smooth muscle trophism. RhoA/Rho-kinase enhance calcium sensitivity and vasoconstriction. Oxidative stress is associated with damage to penile tissues.

Chapter Review

1. Penile erection is dependent on both increased arterial blood flow and coaptation of venous structures.
2. The structure of the corporal bodies of the penis is essential to the erectile process.
3. There is substantial variability in penile vascular anatomy between patients; this may have functional relevance in ED related to pelvic surgery.
4. Neuronal integrity at the brain, spinal cord, and peripheral level are essential to modulation of penile erections.
5. A wide range of neurotransmitters are involved in modulating sexual arousal and penile erection; key examples include dopamine, serotonin, oxytocin, nitric oxide, and norepinephrine.

6. Integration of erotic stimuli in the central nervous system is complex; magnetic resonance imaging (MRI) and positron emission tomography (PET) scanning has provided new insights on central mechanisms of sexual arousal in men.
7. Relaxation of the cavernous smooth muscle is the key to penile erection.
8. NO released by nNOS contained in the terminals of the cavernous nerve initiates the erection process whereas NO released from eNOS in the endothelium helps maintain erection.
9. On entering the smooth muscle cells, NO stimulates the production of cGMP by action on guanylate cyclase.
10. cGMP activates protein kinase G (PKG), which opens potassium channels and closes calcium channels.
11. Low cytosolic calcium favors smooth muscle relaxation.
12. Smooth muscle tone is restored when cGMP is degraded by phosphodiesterase (PDE) and intracellular calcium levels return to baseline levels.
13. Aging is an important contributing factor to ED. The aging process can affect the central regulation of erection, sensory mechanisms, hormonal and neural function, vascular flow, and penile structures.
14. ED is associated with a number of chronic health conditions and may be a sentinel event for serious disease.
15. Diabetes mellitus and metabolic syndrome are associated with particularly severe ED phenotypes. These disease processes may affect multiple organ systems and cause premature aging of central and peripheral structures and molecules that regulate erectile function.
16. Diseases that cause chronic renal failure are associated with ED; renal failure itself may contribute to ED, a condition that may persist despite successful renal transplantation.
17. Drugs most commonly associated with ED include antiandrogens, antidepressants, and antihypertensives.
18. Primary (lifelong) ED may be due to psychogenic causes, congenital arterial insufficiency, penile anomalies, and/or defects of penile vascular structures.

69

Evaluation and Management of Erectile Dysfunction

Arthur L. Burnett II , and Ranjith Ramasamy

Keywords

erectile dysfunction; intracavernosal injection; PDE5 inhibitors; shared decision making; testosterone replacement

Questions

1. The best predictor for the development of erectile dysfunction (ED) is:
    a. young age.
    b. higher education.
    c. underweight status.
    d. passive cigarette smoke exposure.
    e. prediabetes.
2. “Shared decision making” is justified in the modern management of ED because of:
    a. availability of etiology-specific diagnostic testing.
    b. availability of increasingly invasive therapies.
    c. availability of irreversible therapies.
    d. central role of the clinician in directing proper treatment.
    e. acceptance of treatment preferences by patient and partner.
3. The risk association of ED with cardiovascular events:
    a. has not been established.

b. is established unidirectionally to inform ED risk.
c. is established unidirectionally to inform cardiovascular disease risk.
d. is established bidirectionally to inform ED and cardiovascular disease risks.
e. is established in conjunction with carcinogenesis risk.

4. Recommendations for lifestyle modification for the patient with ED require:
   a. stratification of cardiovascular risk.
   b. determination of high and moderate cardiovascular disease risk.
   c. cardiologic workup with diagnostic stress testing standardly.
   d. cardiologic medical treatment.
   e. basic medical assessment and health monitoring.
5. The characteristic that is typically associated with "organic" ED, in contrast with "psychogenic" ED, is:
   a. sudden loss of erections.
   b. situational erectile difficulty.
   c. erection responses on awakening.
   d. gradual decline in erectile ability.
   e. existence of an interpersonal relationship factor.
6. Self-administered ED questionnaires serve to:
   a. document responsiveness to treatment of ED.
   b. define the etiology of ED.
   c. indicate the objective severity of ED.
   d. distinguish the systemic type of ED (e.g., vascular, neurologic, endocrinologic).
   e. evaluate ED in research settings alone.
7. The best rationale for using specialized diagnostic testing is:
   a. to assess complex ED clinical presentations.
   b. as a requirement to direct ED treatment decisions.
   c. to ascertain a specific diagnosis underlying the ED presentation.
   d. to determine ED causation.
   e. to apply Grade A–level evidence-based ED diagnostic tools.

8. Duplex ultrasonography of the penis is a reliable diagnostic test when:
    a. combining pharmacostimulation.
    b. assessing intrapenile vascular communications.
    c. combining intracavernous pressure measurements.
    d. assessing penile and suprapubic penile flow velocities.
    e. indexing results to brachial systolic blood pressures.
9. A vascular test required for a patient under consideration for penile revascularization surgery is:
    a. combined intracavernosal injection and stimulation.
    b. duplex ultrasonography.
    c. dynamic infusion cavernosometry and cavernosography.
    d. penile angiography.
    e. radioisotopic penography.
10. Increased sex hormone–binding globulin (SHBG) is associated with:
    a. elevated bioavailable testosterone.
    b. lowered bioavailable testosterone.
    c. unaltered bioavailable testosterone.
    d. elevated total testosterone.
    e. lowered total testosterone.
11. The efficacy of testosterone therapy is best judged by:
    a. restored diurnal pattern of hormone levels.
    b. normal reference range serum testosterone level.
    c. midrange serum testosterone level.
    d. normal score using hypogonadism questionnaires.
    e. hypogonadism symptomatic improvement.
12. At the penile tissue level, a pharmacologic mechanism to promote penile erection is via the promoting actions of:
    a. cyclic nucleotides.
    b. phosphodiesterases.
    c. α_1 adrenergic agonists.
    d. dopaminergic D_2 receptor agonists.
    e. serotonergic receptor antagonists.

13. Intracavernosal pharmacotherapy is contraindicated for a patient with a clinical history of:
    a. neurological condition.
    b. cardiovascular disease.
    c. diabetes.
    d. priapism.
    e. anticoagulant use.
14. An advantage of alprostadil among pharmacologic agents for intracavernosal pharmacotherapy is:
    a. lower incidence of prolonged erection.
    b. lower incidence of painful erection.
    c. lower cost.
    d. long-term half-life once reconstituted.
    e. role in combination therapy.
15. The vacuum erection device is most advantageous for ED associated with:
    a. veno-occlusive dysfunction.
    b. glanular insufficiency.
    c. postpriapism.
    d. postexplantation of a penile prosthesis.
    e. Peyronie disease.
16. In patients with ED, the vascular lesion addressed by arterial revascularization surgery is:
    a. internal pudendal artery stenosis.
    b. penile dorsal artery stenosis.
    c. cavernosal artery stenosis.
    d. penile deep dorsal venous incompetence.
    e. internal pudendal venous incompetence.

Answers

1. **d. Passive cigarette smoke exposure. Predictors for the development of ED include older age, lower education, diabetes, cardiovascular disease (such as hypertension and**

stroke), cigarette smoke exposure (active and passive), and overweight condition.

2. **e. Acceptance of treatment preferences by patient and partner.** Shared decision making serves to allow the patient or couple to make an informed selection of the preferred therapy for sexual fulfillment on understanding all treatment options after completing a thorough discussion with the treating clinician.
3. **d. Is established bidirectionally to inform ED and cardiovascular disease risks. Epidemiologic studies have documented a bidirectional risk relationship for ED and cardiovascular disease.** This bidirectional paradigm carries ramifications with regard to overall male health status.
4. **e. Basic medical assessment and health monitoring.** All patients presenting for ED management carry some level of cardiovascular risk. Lifestyle modifications such as increased physical activity and improved weight control apply to all patients in accordance with a full medical assessment and regular health monitoring.
5. **d. Gradual decline in erectile ability.** "Organic" ED typically is characterized by a gradual onset of the problem, along with incremental progression, global dysfunction, and poor/absent erections on awakening.
6. **a. Document responsiveness to treatment of ED.** Self-administered ED questionnaires have served to document the presence, subjective severity, and responsiveness of treatment of ED in both clinical and research settings. They do not distinguish an etiologic basis for ED, differentiate among various causes of ED, or indicate an objective severity of ED.
7. **a. To assess complex ED clinical presentations.** Specialized diagnostic testing offers to improve diagnostic accuracy, but it is not a standard requirement to proceed therapeutically. Its best role at this time applies to the ED specialist who may elect certain tests to be done in settings of complex clinical presentations.
8. **a. Combining pharmacostimulation.** The combination of pharmacostimulation or combined intracavernosal injection and stimulation to duplex ultrasonography validates the imaging component of this test by establishing hemodynamic properties of a

functionally relevant erection response. The quantification of blood flow in the penis applies to the main vascular tributaries and includes the entire penis from the crura in the perineum to the tip.

9. **d. Penile angiography.** Definition of the anatomy and radiographic appearance of the iliac, internal pudendal, and penile arteries by penile angiography is necessary in order to perform penile revascularization surgery.
10. **b. Lowered bioavailable testosterone.** Bioavailable testosterone can be affected to some extent by alterations in the SHBG fraction in serum associated with factors that increase SHBG, thereby accounting for a decrease in bioavailable testosterone.
11. **e. Hypogonadism symptomatic improvement.** The objective of testosterone therapy for hypogonadism is symptomatic improvement. The efficacy is not judged by a precise testosterone determination, although standard practice is to provide therapy at a normative serum testosterone level.
12. **a. Cyclic nucleotides. Cyclic guanosine monophosphate and cyclic adenosine monophosphate act to promote proerectile molecular mechanisms resulting in corporal smooth muscle relaxation.**
13. **d. Priapism. Intracavernosal pharmacotherapy is contraindicated for men with psychological instability, a history or risk for priapism, histories of severe coagulopathy or unstable cardiovascular disease, reduced manual dexterity, and use of monoamine oxidase inhibitors.**
14. **a. Lower incidence of prolonged erection.** Perceived advantages of alprostadil for intracavernosal pharmacotherapy relative to other agents are lower incidences of prolonged erection, systemic side effects, and penile fibrosis. Disadvantages include a higher incidence of painful erection, higher cost, and shortened half-life once reconstituted.
15. **b. Glanular insufficiency.** The effect of vacuum erection device therapy involves engorgement of the entire penis including the glans penis, such that it provides an advantage to patients experiencing glanular insufficiency. Special uses for this therapy such as preserving the elasticity of penile tissue after priapism or

penile prosthesis explantation or after surgical correction of Peyronie disease have been suggested.

16. **a. Internal pudendal artery stenosis.** The penile arterial anatomic defect correctable by arterial revascularization commonly involves stenosis of the internal pudendal artery following perineal or pelvic trauma.

Chapter Review

1. The prevalence of ED in the adult male is 20% and is correlated with age.
2. Men with ED are 45% more likely than men without ED to experience a cardiac event within 5 years of diagnosis.
3. The International Index of Erectile Function (IIEF) questionnaire has five domains: erectile function, orgasmic function, sexual desire, intercourse satisfaction, and overall satisfaction.
4. Erections observed with nocturnal penile tumescence monitoring do not necessarily equate with erections sufficient for sexual performance.
5. Cavernous arterial insufficiency is suggested when peak systolic velocity is less than 25 cm/s. A peak systolic velocity greater than 35 cm/s defines normal cavernous arterial inflow.
6. Testosterone circulates as free, bound to albumin, and bound to SHBG. Free testosterone and albumin-bound testosterone comprise the bioavailable testosterone.
7. Testosterone production is circadian, with the peak occurring in the morning. To evaluate testosterone status, blood should be drawn in the morning (between 7 AM and 11 AM).
8. Penile revascularization is reserved for patients in whom it is most likely to be successful: patients with a history of perineal trauma, age younger than 55 years, nondiabetic, nonsmoker, no venous leak, and a documented stenotic lesion in the internal pudendal artery on angiography.
9. Endocrine conditions that may be associated with erectile dysfunction include hypogonadism, hyperthyroidism, and diabetes.

10. Induction of a penile erection requires release of nitric oxide from penile nerve endings and vascular endothelium.
11. Nitrate use in any form is an absolute contraindication for the use of phosphodiesterase type 5 (PDE5) inhibitors.
12. Penile rehabilitation following radical prostatectomy using PDE5 inhibitors has not proved efficacious.
13. The vasoactive drugs commonly injected to produce an erection include prostaglandin E, papaverine, and phentolamine.
14. Predictors for the development of ED include older age, lower education, diabetes, cardiovascular disease (such as hypertension and stroke), cigarette smoke exposure (active and passive), and overweight condition.
15. Intracavernosal pharmacotherapy is contraindicated for men with psychological instability, a history or risk for priapism, histories of severe coagulopathy or unstable cardiovascular disease, reduced manual dexterity, and use of monoamine oxidase inhibitors.

70

Priapism

Gregory A. Broderick

Keywords

high-flow priapism; ischemic priapism; priapism; recurrent ischemic priapism; sickle cell disease; stuttering priapism

Questions

1. Ischemic priapism is a persistent erection marked by each of the following clinical and pathophysiologic characteristics EXCEPT:
    a. rigidity of the corpora cavernosa.
    b. bright red corporal blood.
    c. hypoxic and acidotic corporal environment.
    d. painful rigidity.
    e. thrombus within the sinusoidal spaces.
2. Each of the following are etiologies typically associated with ischemic priapism EXCEPT:
    a. sickle cell disease (SCD).
    b. straddle injury.
    c. cocaine use.
    d. spider bite.
    e. pharmacologic erection therapy.
3. SCD is a risk factor for ischemic priapism; the pathophysiologic mechanisms include each of the following EXCEPT:
    a. decreased content of hemoglobin S (HgbS) in the plasma.

b. the polymerization of HgbS when deoxygenated.
c. scavenging of nitric oxide (NO).
d. arginine catabolism removing substrate for NO synthesis.
e. adhesive interactions among sickle cells, endothelial cells, and leukocytes.

4. Prolonged erection in males 40 years of age and older is usually attributed to:
   a. SCD.
   b. hematologic malignancy.
   c. erectile dysfunction (ED) pharmacotherapy.
   d. prostate cancer.
   e. testosterone supplementation.
5. Case reports have documented prolonged erections and, rarely, priapism in men by using phosphodiesterase type 5 (PDE5) inhibitor therapies. Associated risks for prolonged erection/priapism include each of the following EXCEPT:
   a. daily dosing.
   b. combination with intracavernous injection.
   c. history of penile trauma.
   d. psychotropic medications.
   e. narcotic use.
6. The associations and pathophysiology of high-flow priapism include each of the following EXCEPT:
   a. straddle injury.
   b. coital trauma.
   c. birth canal injury to the newborn male.
   d. cold-knife urethrotomy.
   e. hemodialysis.
7. The critical pathologic change occurring in the cavernosal tissue at 4 hours after the onset of ischemic priapism is:
   a. irreversible cavernous damage and ED.
   b. the beginning of glucopenia.
   c. the beginning of hypercoagulable thrombotic conditions.
   d. the deterioration of cavernous smooth muscle contractile responses.
   e. cavernous fibrosis.

8. The NO/cyclic guanosine monophosphate (cGMP) signaling pathway is implicated in the pathogenesis of priapism on the basis of scientific work showing:
    a. guanylate cyclase activity upregulation.
    b. guanylate cyclase activity downregulation.
    c. NO synthase activity upregulation.
    d. PDE5 activity upregulation.
    e. PDE5 activity downregulation.
9. An adolescent with SCD presents with a 6-hour erection. Initial cavernous blood gas results show P_{O_2} 30 mm Hg, P_{CO_2} 60 mm Hg, and pH 7.25. The first therapeutic step should be:
    a. oral terbutaline.
    b. oral pseudoephedrine.
    c. intracavernous aspiration.
    d. exchange transfusion.
    e. distal surgical shunt.
10. The characteristic blood flow defect of ischemic priapism found on color duplex ultrasonography is:
    a. normal cavernosal artery inflow.
    b. increased cavernosal artery inflow.
    c. decreased or absent cavernosal artery inflow.
    d. arteriovenous blush.
    e. sinusoidal fistula.
11. After initial intracavernous treatment for ischemic priapism, blood gas sampling produces an equivocal mixed-venous blood result. Priapism resolution is best confirmed by:
    a. color duplex ultrasonography.
    b. penile scintigraphy.
    c. corpus cavernosography.
    d. penile arteriography.
    e. pelvic computed tomography scan.
12. After a second session of intracavernous treatment consisting of aspiration/irrigation with phenylephrine administration, the priapic penis remains turgid. Cavernosal blood gas results are P_{O_2} 40 mm Hg, P_{CO_2} 50 mm Hg, and pH 7.35. The next step should be:

a. observation.
b. oral sympathomimetic.
c. repeat intracavernous treatment.
d. distal surgical shunt.
e. proximal surgical shunt.

13. Phenylephrine is the preferred sympathomimetic used in the treatment of ischemic priapism because of its:
    a. α_1-selective activity.
    b. α_1 and α_2 activity.
    c. β_1-selective activity.
    d. β_2-selective activity.
    e. combined α and β activities.
14. The best indication for arterial embolization in the management of high-flow priapism is:
    a. unlikely spontaneous resolution.
    b. failure of sympathomimetic therapy.
    c. reduction of recurrent priapism risk.
    d. reduction of subsequent ED risk.
    e. patient preference to intervene.
15. Persistent penile rigidity after a technically successful proximal surgical shunt procedure in a patient with a 72-hour episode of ischemic priapism is an indication for:
    a. observation.
    b. gonadotropin-releasing hormone agonist therapy.
    c. pudendal artery ligation.
    d. distal surgical shunt.
    e. penile prosthesis surgery.
16. The mother of a child with SCD complains that her son has recently been awakening with erections lasting 3 to 4 hours. She is concerned that similar occurrences have been a warning sign for a major priapism. All of the following are appropriate management options EXCEPT:
    a. trial of nightly oral sympathomimetic drug.
    b. trial of low-dosage, daily PDE5 inhibitor.
    c. a gonadotropin-releasing hormone agonist or antiandrogen.

d. intracavernous injection of phenylephrine in the morning.

17. Evidence-based studies of priapism therapies and outcomes are rare. A recent investigation of adult SCD patients presenting with ischemic priapism subjected all men to a standard protocol of aspiration and phenylephrine injections. Long-term sexual health function outcomes revealed complete ED in men with duration of priapism:
    a. less than 12 hours.
    b. 12 to 24 hours.
    c. longer than 36 hours.
    d. longer than 48 hours.
    e. longer than 72 hours.
18. An adult male presents with ischemic priapism of 8 hours duration. He fails to respond to serial aspiration and intracavernous injection after 4 hours in the emergency department. The recommended intervention at this time should be:
    a. hydration, nasal oxygen, and keeping the patient NPO for 8 hours to avoid risks of emergent intubation.
    b. a percutaneous distal penile shunt.
    c. an open distal shunt.
    d. an open proximal shunt.
    e. a saphenous vein shunt.
19. Radiographic imaging may be helpful in the diagnosis and management of priapism. Each of the following is correct EXCEPT:
    a. color Doppler ultrasound (CDU) in the evaluation of a persistent erection following treatments for ischemic priapism.
    b. penile arteriography to differentiate high-flow from ischemic priapism.
    c. magnetic resonance imaging (MRI) to diagnose corporal thrombus in men with refractory priapism or when there has been a significant delay in presentation.
    d. MRI in the differential diagnosis of corporal metastasis.
20. A 36-year-old tech entrepreneur is referred for a diagnosis of priapism after he slipped while climbing aboard his yacht. He initially had a saddle bruise on his perineum and pain; the next

morning he awoke with persistent erection. The patient has a board of directors meeting at the end of the week and wants immediate treatment. The correct management strategy is:

a. penile aspiration and α-adrenergic injection in the office or emergency department.
b. penile arteriography.
c. angio-embolization after a thorough discussion of chances for spontaneous resolution and risks of treatment-related ED.
d. CDU-guided corporal exploration to ligate fistula.
e. distal penile shunt.

21. Priapism associated with SCD is ischemic. The current pathophysiology is believed to be:
    a. obstruction of venous outflow by sickled erythrocytes.
    b. hemolysis and reduced NO.
    c. increased blood viscosity.
    d. a blood dyscrasia associated with reduced reticulocyte counts.
    e. dysregulated cavernous arterial inflow.
22. Ischemic priapism in boys and men with SCD should focus on correcting the hemoglobinopathy:
    a. with exchange transfusions.
    b. with hydration, alkalinization, and oxygen.
    c. with aspiration and pharmacologic detumescence.
    d. with hydroxyurea.
    e. with analgesia and expectant management.
23. A 27-year-old previously healthy white male is brought into the emergency department in the early morning hours by his girlfriend. She describes a night of partying with alcohol, energy drinks, and a small amount of cocaine. Her boyfriend has had a persistent erection for 6 to 8 hours. She says, “I’ve done all I can to help him —it won’t go down!” The emergency department doctor calls the urologist to go over the case and initiate management. The best course of therapy would be:
    a. baclofen, 20 mg PO followed every 6 hours until erection subsides.
    b. pseudoephedrine, 60 mg PO.
    c. inhaled terbutaline, 4 mg.

d. oral terbutaline, 4 mg.
e. corporal aspiration and injection of phenylephrine, 200 µg.

24. PDE5 inhibitors are first-line therapy for ED, but like intracavernous therapy, PDE5 inhibitors have been associated with prolonged erection and priapism. Which agent is most likely to cause priapism?
    a. Sildenafil, 100 mg PRN sex
    b. Avanafil, 200 mg PRN sex
    c. Levitra, 20 mg PRN sex
    d. Cialis, 5 mg daily
    e. None of these agents is more likely to result in priapism
25. Early reviews concluded that the natural history of ischemic priapism is ED. Recent interventions have tracked sexual health function outcomes with evidence-based questionnaires such as the International Index of Erectile Function (IIEF-5). Improvements in medical and surgical management of ischemic priapism may preserve erectile function. Unfortunately, ED is likely to occur if reversal by medical or surgical means is not successful after how many hours?
    a. 8 hours
    b. 12 to 24 hours
    c. 36 hours
    d. 48 hours
    e. 72 hours

Answers

1. **b. Bright red corporal blood.** Ischemic priapism is a persistent erection marked by rigidity of the corpora cavernosa and little or no cavernous arterial inflow. **In ischemic priapism there are time-dependent changes in the corporal metabolic environment with progressive hypoxia, hypercarbia, and acidosis. The patient typically complains of penile pain after 6 to 8 hours, and examination reveals a rigid erection.** After 48 hours, thrombus

can be found in the sinusoidal spaces and smooth muscle necrosis with fibroblast-like cell transformation is evident.

2. **b. Straddle injury. Nonischemic priapism is much rarer than ischemic priapism, and the etiology is largely attributed to trauma.** Forces may be blunt or penetrating, resulting in laceration of the cavernous artery or one of its branches within the corpora. The etiology most commonly reported is a straddle injury to the crura.
3. **a. Decreased content of hemoglobin S (HgbS) in the plasma.** The clinical features are seen in homozygous SCD patients: chronic hemolysis, vascular occlusion, tissue ischemia, and end-organ damage. HgbS polymerizes when deoxygenated, injuring the sickle erythrocyte, activating a cascade of hemolysis and vaso-occlusion. Membrane damage results in dense sickling of red cells, causing adhesive interactions among sickle cells, endothelial cells, and leukocytes. Hemolysis releases hemoglobin into the plasma. Free Hgb reacts with NO to produce methemoglobin and nitrate. **This is a scavenging reaction; the vasodilator NO is oxidized to inert nitrate.** Sickled erythrocytes release arginase-I into blood plasma, which converts L-arginine into ornithine, effectively removing substrate for NO synthesis. Oxidant radicals further reduce NO bioavailability. **The combined effects of NO scavenging and arginine catabolism result in a state of NO resistance and insufficiency termed *hemolysis-associated endothelial dysfunction.*** Therapeutic interventions include transfusion to decrease the relative concentrations of HgbS in the plasma.
4. **c. Erectile dysfunction (ED) pharmacotherapy.** The introduction of intracavernous pharmacotherapy approximately two decades ago led to a pronounced increase in the incidence of prolonged erection and true priapism. Prolonged erection is more commonly reported than is priapism, following therapeutic or diagnostic injection of intracavernous vasoactive medications. In many communities patients receiving intracavernous medications for ED will outnumber patients with SCD. The majority of men requiring treatment for ED are middle-aged to older men. In worldwide clinical trials of the Alprostadil Study Group, prolonged erection

(defined as 4 to 6 hours) was 5%, and priapism (> 6 hours) was described in 1% of subjects. In papaverine/phentolamine/alprostadil intracavernous injection programs, prolonged erections have been reported in 5% to 35% of patients.

5. **a. Daily dosing.** Few cases reports have documented priapism following PDE5 inhibitor therapy. **These reports suggest that men with the following conditions were at increased risk for priapism: SCD, spinal cord injury, men who used a PDE5 inhibitor recreationally, men who used a PDE5 inhibitor in combination with intracavernous injection, men with a history of penile trauma, men on psychotropic medications, and men abusing narcotics.** Tadalafil, 5 mg daily dosing caused no priapism in a phase II clinical study of 281 men with history of lower urinary tract symptoms secondary to benign prostatic hyperplasia (BPH) for 6 weeks, followed by dosage escalation to 20 mg once daily for 6 weeks.
6. **e. Hemodialysis. Nonischemic priapism is much rarer than ischemic priapism, and the etiology is largely attributed to trauma. Forces may be blunt or penetrating, resulting in laceration of the cavernous artery or one of its branches within the corpora.** The etiology most commonly reported is a straddle injury to the crura. Other mechanisms include coital trauma, kicks to the penis or perineum, pelvic fractures, birth canal trauma to the newborn male, needle lacerations, complications of penile diagnostics, and vascular erosions complicating metastatic infiltration of the corpora. Although accidental blunt trauma is the most common etiology, high-flow priapism has been described following iatrogenic injury: cold-knife urethrotomy, Nesbitt corporoplasty, and deep dorsal vein arterialization. Any mechanism that lacerates a cavernous artery or arteriole can produce unregulated pooling of blood in sinusoidal space with consequent erection.
7. **d. The deterioration of cavernous smooth muscle contractile responses.** In ischemic priapism there are time-dependent changes in the corporal metabolic environment with progressive hypoxia, hypercarbia, and acidosis. In vitro studies have demonstrated that

when corporal smooth muscle strips are exposed to a hypoxic condition, α-adrenergic stimulation fails to induce corporal smooth muscle contraction. Histologically, by 12 hours corporal specimens show interstitial edema, progressing to destruction of the sinusoidal endothelium, exposure of the basement membrane, and thrombocyte adherence at 24 hours.

8. **e. PDE5 activity downregulation. Recent scientific advances have shown that priapism is associated with decreased PDE5 functional regulation in the penis.** The relative lack of this molecular factor needed for controlling chemical signaling of penile erection accounts for stuttering and ischemic priapism in sickle cell patients. Experimental models of corpus cavernosum smooth muscle cells suggest that the cyclic nucleotide cGMP is produced in low steady-state amounts under the influence of priapism-related destruction of the vascular endothelium and thus reduced endothelial NO activity. This situation downregulates the set point of PDE5 function, secondary to altered cGMP-dependent feedback control mechanisms. When NO is neuronally produced in response to erectogenic stimulus or during sleep-related erectile activity, cGMP production surges in a manner that leads to excessive erectile tissue relaxation because of basally insufficient functional PDE5 to degrade the cyclic nucleotide.
9. **c. Intracavernous aspiration.** The history and blood gases define an ischemic priapism event, which warrants immediate attempts at decompression/aspiration of the corpora cavernosa. **In SCD, concurrent systemic treatments may be offered, but relief of the penile ischemia should be pursued aggressively. Aspiration should be repeated until no more dark blood can be seen coming out of the corpora and bright red blood is obtained.** This process decreases the intracavernous pressure, relieves pain, and resuscitates the corporal environment, removing anoxic, acidotic, and hypercarbic blood.
10. **c. Decreased or absent cavernosal artery inflow.** In presentations of ischemic priapism, minimal or absent blood flow in the cavernosal arteries is found using color Doppler ultrasonography. CDU is an adjunct to the corporal aspirate in differentiating

ischemic from nonischemic priapism. Patients with ischemic priapism will have no blood flow in the cavernous arteries; the return of the cavernous artery waveform will accompany successful detumescence. Patients with nonischemic priapism have normal to high blood-flow velocities in the cavernous arteries; an effort should be made to localize the characteristic blush of color emanating from flow signal of the disrupted cavernous artery or arteriolar-sinusoidal fistula.

11. **a. Color duplex ultrasonography.** This tool is the most reliable and least invasive imaging technique to differentiate ischemic from nonischemic priapism. **CDU should always be considered in the evaluation of a full or partial erection after treatments for ischemic priapism. The differential diagnosis includes resolved ischemia with persistent tenderness and penile edema, persistent ischemia, or conversion to a high-flow state.**
12. **a. Observation.** These blood gas results are consistent with normal mixed venous blood. The turgid penis may be due to tissue edema. Observation is appropriate at this time.
13. **a. α_1-Selective activity.** According to American Urological Association Guidelines, aspiration followed by intracavernous injection of sympathomimetic drugs is the standard of care in the medical treatment of ischemic priapism. Sympathomimetic pharmacotherapies (phenylephrine, etilefrine, ephedrine, epinephrine, norepinephrine, metaraminol) cause smooth muscle contraction in the corpora. Phenylephrine is a selective α_1-adrenergic receptor agonist, with minimal β-mediated inotropic and chronotropic cardiac effects. There are no comparative trials of sympathomimetic injectables in the management of priapism.
14. **e. Patient preference to intervene. Nonischemic priapism should generally be managed by observation.** Arterial priapism is not an emergency. **Spontaneous resolution or response to conservative therapy has been reported in up to 62% of published series.** Persistent partial erection from high-flow priapism may continue from months to years. There are no comparative outcome studies of intervention versus conservative management, but there are

sufficient case descriptions in children to recommend initial watchful waiting. Adult patients demanding immediate relief can be offered selective arterial embolization.

15. **e. Penile prosthesis surgery. Unfortunately, the natural history of untreated ischemic priapism or priapism refractory to interventions is severe fibrosis, penile length loss, and complete ED.** The exact time point when penile prosthesis becomes a reasonable option is unclear, but most series describe complete ED in men who have had ischemic priapism for 36 to 72 hours. The advantages of early penile implantation in the acute management of ischemic priapism are preservation of penile length and a technically easier implantation; delayed implantation is technically challenging due to corporal fibrosis.
16. **c. A gonadotropin-releasing hormone agonist or antiandrogen.** The goals of managing stuttering ischemic priapism are prevention of future episodes, preservation of erectile function, and reducing the trauma to the patient from priapism management. A trial of daily oral sympathomimetic therapy, a trial of oral PDE5 inhibitor therapy, or intracavernous injection of phenylephrine should be considered in the management of children and adults with stuttering ischemic priapism associated with hemoglobinopathies. **GnRH agonists or antiandrogens may be used in adults but should not be used in patients who have not achieved full sexual maturation and adult stature.**
17. **c. Longer than 36 hours.** In those patients in whom priapism was reversed, spontaneous erection (with or without sildenafil) was reported in 100% of men when priapism was reversed in less than 12 hours; 78% when reversed by 12 to 24 hours; and 44% when reversed by 24 to 36 hours. **No patient reported spontaneous erection after priapism duration of more than 36 hours.**
18. **b. A percutaneous distal penile shunt.** The objective of shunt surgery is reoxygenation of the corpus cavernosum. **The shared principle of all shunting procedures is to reestablish corporal inflow by relieving venous outflow obstruction. A distal cavernoglanular shunt should be the first choice of shunting procedures because it is technically easier to perform than**

proximal shunting. Percutaneous distal shunting is less invasive than open distal shunting and can be attempted in the emergency department with local anesthetic. Anesthesiologists must be educated that ischemic priapism is an emergency and sexual function outcomes are time dependent; appropriate airway precautions should be taken for emergent intubation as needed in the surgical management of priapism.

19. **b. Penile arteriography to differentiate high-flow from ischemic priapism.** CDU is an adjunct to the corporal aspirate in differentiating ischemic from nonischemic priapism. CDU imaging should include corporal shaft and transperineal assessment of the crural bodies when there is a history of penile trauma or straddle injury. CDU should always be considered in the evaluation of a persistent or partial erection after treatments for ischemic priapism. Penile arteriography is too invasive as a diagnostic procedure to differentiate ischemic from nonischemic priapism. Penile arteriography should be reserved for the management of high-flow priapism, when embolization is planned. There are three possible roles for MRI: (1) imaging of a well-established arteriolar-sinusoidal fistula, (2) identifying corporal thrombus, and (3) identifying corporal metastasis.
20. **c. Angio-embolization after a thorough discussion of chances for spontaneous resolution and risks of treatment-related ED.** Arterial priapism is not an emergency and may be managed expectantly. Diagnosis of high-flow priapism is best made by penile/perineal CDU. Penile aspiration and injection of α-adrenergic agents is not recommended for HFP. Angio-embolization should be preceded by a thorough discussion of chances for spontaneous resolution, risks of treatment-related ED, and lack of significant consequences expected from delaying interventions. Overall success rates with embolization are high, although a single treatment carries a recurrence rate of 30% to 40%. **Where angio-embolization fails or is contraindicated, surgical ligation is reasonable. Formation of a pseudocapsule at the site of a sinusoidal fistula may take weeks to months**

following trauma. CDU guidance is recommended during exploration to locate the fistula.

21. **b. Hemolysis and reduced NO. Contemporary science implicates hemolysis and reduced NO in the pathogenesis of pulmonary hypertension, leg ulcers, priapism, and stroke in SCD patients, whereas increased blood viscosity is believed to be responsible for painful crises, osteonecrosis, and acute chest syndrome**. SCD patients with priapism have a fivefold greater risk of developing pulmonary hypertension. SCD priapism is also associated with reduced hemoglobin levels and increased hemolytic markers: reticulocyte count, bilirubin, lactate dehydrogenase, and aspartate aminotransferase.
22. **c. With aspiration and pharmacologic detumescence.** Classically, management of SCD-induced ischemic priapism involved analgesics, hydration, oxygen, bicarbonate, and exchange transfusion. It is very tempting, especially in boys, to defer management to pediatricians and hematologists. Hematologists have begun to question the emphasis on intravenous hydration, sodium bicarbonate for alkalinization, and exchange transfusion as first-line therapy for SCD-associated priapism. Unfortunately, acute neurologic complications may follow exchange transfusions. Hydroxycarbamide (hydroxyurea) is a hematologic agent used in the management of vaso-occlusive crises in SCD patients. The proposed mechanisms of action are increase in production of hemoglobin F, lowering of leukocytes–platelets–reticulocytes, and promoting release of NO. In the best interest of the patient, the urologist should seek hematologic consultation in the management of boys and men with SCD priapism, but remain assertive that hematolog ic therapy alone is not effective management of SCD priapism. One report suggested that blood transfusion may have no effective role in the treatment of sickle cell–induced priapism. Reports from hematology centers suggest high success rates using penile aspiration/injection/irrigation of intracavernous sympathomimetics for SCD priapism.
23. **e. Corporal aspiration and injection of phenylephrine 200 μg.** Oral agents are not recommended in the management of acute

ischemic priapism (> 4 hours). Oral α-adrenergic agents may have a role in the management of stuttering priapism associated with prolonged nocturnal erections. The recommended initial treatment of ischemic priapism is the decompression of the corpora cavernosa by aspiration. Aspiration will immediately soften the erection and relieve pain. Aspiration alone may relieve priapism in 36% of cases. The AUA Guidelines Panel advised that there were not sufficient data to conclude that aspiration followed by saline intracorporal irrigation was any more effective than aspiration alone. Aspiration should be repeated until no more dark blood can be seen coming out from the corpora and fresh bright red blood is obtained. This process leads to a marked decrease in the intracavernous pressure, relieves pain, and resuscitates the corporal environment removing anoxic, acidotic, and hypercarbic blood. **Corporal aspiration, if unsuccessful, should be followed by α-adrenergic injection.**

24. **e. None of these agents is more likely to result in priapism.** The 2013 label for the most recently approved PDE5 inhibitor Stendra (avanafil 50 mg, 100 mg, 200 mg) contains precautionary wording virtually identical to that on prior labels for PRN oral dosing of sildenafil, vardenafil, and tadalafil: "There have been rare reports of prolonged erection greater than 4 hours and priapism (painful erections greater than 6 hours)." Once-daily tadalafil (2.5 mg and 5 mg) was approved for oral treatment of ED in 2008, and subsequently in 2011 tadalafil (2.5 mg and 5 mg) was approved for the signs and symptoms of BPH and treatment of ED and the signs and symptoms of BPH. Tadalafil 5 mg daily dosing caused no priapism in a phase II clinical study of 281 men with history of lower urinary tract symptoms secondary to BPH for 6 weeks.
25. **d. 48 hours.** Bennett and Mulhall in 2008 carefully documented 39 cases of SCD priapism presenting to their emergency department over 8 years; men were routinely interviewed for erectile function status within 4 weeks of priapism/interventions. Of the 39 African-American men followed, 73% acknowledged prior episodes of stuttering; 85% had previously been diagnosed with SCD; but only 5% had been counseled in SCD clinics or were aware that priapism

was a complication of SCD. A standard protocol of aspiration and phenylephrine injection was performed; shunting for failure of medical management was performed in 28%. In those patients in whom priapism was reversed, spontaneous erections (with or without use of sildenafil) were reported in 100% of men when priapism was reversed by 12 hours; 78% when reversed by 12 to 24 hours; and 44% when reversed by 24 to 36 hours. In this contemporary series of SCD patients, no men reported the return of spontaneous erections after priapism lasting 36 hours or more. Ralph describes the efficacy and outcomes of combining the T-shunt with the corporal "snake" maneuver in 45 patients. All were refractory to medical reversal of ischemic priapism. The combined distal surgical technique was successful in resolving the acute priapism if duration was less than 24 hours but had limited efficacy in cases of priapism greater than 48 hours. Corporal needle biopsies were performed in each case and documented smooth muscle necrosis, worsening as a function of time and uniform in all men with more than 48 hours of ischemia. At 6 months erectile function outcomes were assessed by the erectile function domain score from the IIEF-5. T-shunt with corporal snake tunneling successfully reversed ischemic priapism in all patients with less than 24 hours' duration, but at 6 months ED was complained of by 50% of men. **The authors conclude that the cutoff for irreversible restoration of erectile tissue is 48 hours; aggressive combined distal shunting may resolve priapism in a small cohort (30%), but all will have severe ED. They advise that management of refractory ischemic priapism greater than 48 hours should include discussion of immediate insertion of penile implant.**

Chapter Review

1. Ischemic priapism is the result of little or no cavernous arterial inflow.
2. Interventions 48 hours beyond the onset of ischemic priapism may relieve pain but will have little benefit in preserving potency.

3. Stuttering priapism is recurrent, unwanted, painful erections in men, usually with sickle cell disease.
4. Nonischemic or high-flow arterial priapism is usually the result of trauma, and the corpora are tumescent but not rigid.
5. The etiology of the majority of ischemic priapism is idiopathic; however, it may be associated with psychotropic medications, alcohol or drug abuse, hematologic dyscrasias, metastatic disease, perineal trauma, intracavernous injection of vasoactive drugs, and sickle cell disease.
6. Priapism following a PDE5 inhibitor usually occurs in men with other risk factors.
7. On rare occasions, following reversal of ischemic priapism, high-flow priapism may occur. Doppler ultrasound is useful in making this diagnosis.
8. Ischemic priapism and stuttering priapism are the result of NO imbalance.
9. In ischemic priapism, aspiration alone will be curative in a third of the patients.
10. Phenylephrine, 200 μg/mL given in 0.5-mL aliquots, not to exceed a total dose of 1 mg, is used to treat ischemic priapism. Corporal aspiration, if unsuccessful, should be followed by α-adrenergic injection.
11. Aspiration has no role in high-flow priapism other than for diagnosis; it plays no role in treatment.
12. PDE5 inhibitor therapy has been used to treat stuttering priapism in men with sickle cell disease.
13. When surgical therapy is indicated, a cavernosal glanular shunt should be the first choice.
14. Spontaneous resolution of high-flow arterial priapism generally occurs in two thirds of patients.
15. If a distal shunt fails, a proximal shunt should be considered.
16. In sickle cell disease, concurrent systemic treatments may be offered, but relief of the penile ischemia should be pursued aggressively. Aspiration should be repeated until no more dark blood can be seen coming out of the corpora and bright red blood is obtained.

17. Increased risk for priapism occurs in sickle cell disease, spinal cord injury, men who used a PDE5 inhibitor recreationally, men who used a PDE5 inhibitor in combination with intracavernous injection, men with a history of penile trauma, men on psychotropic medications, and men abusing narcotics.

71

Disorders of Male Orgasm and Ejaculation

Chris G. McMahon

Keywords

anejaculation; delayed ejaculation; ejaculation; ejaculatory anhedonia; orgasm headache;; orgasm; painful ejaculation; post-orgasm illness syndrome; post-SSRI syndrome; premature ejaculation; retrograde ejaculation

Questions

1. The most important inhibitory neurotransmitter in the central neurochemical control of ejaculation is:
    a. dopamine
    b. serotonin
    c. noradrenalin
    d. oxytocin
    e. gamma aminobutyric acid (GABA)
2. Which of the following is incorrect?
    a. The ejaculatory process comprises three phases: emission, ejection, and orgasm.
    b. Emission is synergistic activation of both sympathetic and parasympathetic nerves (T10 to L2) but may also include a cholinergic excitatory mechanism.
    c. Ejection is mediated by somatic nerves (S2 to S4).

d. Ejection involves a sympathetic spinal cord reflex upon which there is limited voluntary control.
e. Closure of bladder neck, rhythmic contraction of the bulbocavernosus, bulbospongiosus, and other pelvic floor muscles, and relaxation of the internal urinary sphincter during emission prevents retrograde ejaculation.

3. The prevalence of intravaginal ejaculation latency times (IELT) of ≤1 minute in the general population is approximately
   a. <1%
   b. 2.5%
   c. 5.5%
   d. 11%
   e. 20%
4. Which of the following is incorrect?
   a. The *Diagnostic and Statistical Manual of Mental Disorders* (DSM-V) definition of premature ejaculation (PE) is an operationalized multivariate definition that captures the key dimensions of latency, control, and bother.
   b. Approximately 50% of heterosexual men seeking treatment for lifelong PE ejaculate within 1 minute after penetration.
   c. The consistent early ejaculations of lifelong PE suggest an underlying neurobiological functional disturbance.
   d. Acquired PE is most often due to psychological factors or comorbid erectile dysfunction (ED).
   e. Men with subjective PE complain of PE, often due to psychological and/or cultural factors, but have a normal or even extended IELT.
5. Which of the following is incorrect?
   a. The International Society for Sexual Medicine (ISSM) developed the first contemporary, evidence-based definition of lifelong and acquired PE.
   b. The main constructs in the ISSM definition of PE are time from penetration to ejaculation, inability to delay ejaculation, and the presence of negative personal consequences.
   c. Lifelong PE is, in part, defined by an IELT less than about 1 minute.

d. Acquired PE is, in part, defined as a clinically significant and bothersome reduction in latency time, often to about 3 minutes or less.
e. This definition is applicable to all men regardless of their sexual orientation or type of sexual contact.

6. The rationale for the ISSM definition of lifelong and acquired PE includes which of the following?
    a. Men with acquired PE tend to have lower IELT than men with lifelong PE
    b. Patient self-estimated IELT correlates poorly with stopwatch IELT
    c. Most men do not use cognitive or behavioral techniques to prolong intercourse and delay ejaculation
    d. Men with lifelong or acquired PE invariably experience a variety of negative psychological consequences such as bother, frustration, or the avoidance of sexual contact
    e. Men with PE had similar overall health-related quality of life, self-esteem, and confidence compared to non-PE groups
7. Which of the following is false?
    a. There is a substantial disparity between the incidence of PE in epidemiological studies that rely upon either patient self-report of PE and/or inconsistent and poorly validated definitions of PE.
    b. Community-based stopwatch IELT studies demonstrate that the distribution of the IELT is positively skewed, with a median IELT of 5.4 minutes (range, 0.55 to 44.1 minutes).
    c. Community-based stopwatch IELT studies demonstrate that IELT decreases with age and varies between countries.
    d. Some men may have a neurobiological and genetic predisposition toward early ejaculation.
    e. Subthyroidism is a common cause of PE.
8. In men with acquired PE and comorbid ED, which of the following is false?
    a. Intentionally “rushing” intercourse to prevent early detumescence of a partial erection is common

b. PE is rarely compounded by the presence of high levels of performance anxiety related to their ED
c. Limitation of arousal during foreplay to prevent early ejaculation may result in an incomplete erection
d. The presence of vascular risk factors such as diabetes and hypertension is common
e. Treatment with a PDE5 inhibitor alone or in combination with an SSRI is regarded as first-line treatment

9. In the evaluation of men with acquired PE, which of the following is false?
    a. A physical examination is mandatory in an effort to identify the etiology of the PE
    b. The presence of comorbid ED can be evaluated using a validated instrument such as the International Index of Erectile Function (IIEF) or the IIEF-5 Sexual Health Inventory for Men (SHIM)
    c. The index of premature ejaculation (IPE) was developed specifically for use as a screening questionnaire
    d. Laboratory or imaging investigations are occasionally required based upon the patient's medical history
    e. A digital prostate examination, routine in an andrological setting for all men over 40, is useful in identifying possible evidence of prostatic inflammation or infection
10. In the management of PE, which of the following is false?
    a. All men seeking treatment for PE should receive basic psychosexual education or coaching.
    b. Present-day psychotherapy for PE is an integration of behavioral (e.g., the well-known start-stop and pause-squeeze methods) and cognitive approaches within a short-term psychotherapy model.
    c. Psychological-behavioral strategies for treating PE are moderately successful in the long term.
    d. Drawbacks to the psychological-behavioral approach are that it is time-consuming, requires substantial resources of both time and money, lacks immediacy, and requires the partner's cooperation.

e. Combining PE pharmacotherapy and psychological approaches may be especially useful and potentially prevent symptom relapses.

11. In the pharmacotherapy management of PE, which of the following is false?
    a. Dapoxetine is a rapid-acting and short half-life SSRI taken 1 to 3 hours before sexual contact.
    b. Daily dosing of selective serotonin reuptake inhibitors (SSRIs) such as paroxetine, sertraline, fluoxetine, and citalopram are effective treatments for PE.
    c. SSRIs inhibit the postsynaptic 5-HT transporter system in the serotonergic neuron synapse.
    d. Dapoxetine can be combined with sildenafil as a treatment for PE in men with comorbid ED.
    e. The use of on-demand clomipramine taken 4 to 6 hours prior to sexual contact is limited by the occurrence of nausea and dizziness.
12. Which of the following is/are true?
    a. The DSM-IV-TR definition of delayed ejaculation is evidence-based on a small number of community-based IELT studies.
    b. Men with IELT greater than 2 standard deviations above the mean who report distress and/or cease sexual intercourse due to fatigue or irritation are regarded as suffering from delayed ejaculation.
    c. An IELT of 40 to 45 minutes represents an IELT in excess of 2 standard deviations above the mean.
    d. Many men with acquired DE can often masturbate to orgasm and may use idiosyncratic masturbation techniques that cannot be easily replicated during intercourse.
    e. Men with delayed ejaculation often have difficulty attaining or keeping their erections.
13. The causes of delayed ejaculation include which of the following? (Select all that apply.)
    a. Hyperthyroidism
    b. Hypogonadism
    c. Hyperprolactinemia

d. Multiple sclerosis
e. Antidepressant drugs such as SSRIs

14. In men with a spinal cord injury (SCI), which of the following is false? (Select all that apply.)
    a. The ability to ejaculate increases with descending levels of spinal injury**.**
    b. Less than 5% of patients with complete upper motor neuron lesions retain the ability to ejaculate.
    c. The ability to achieve an erection increases with descending levels of spinal injury.
    d. Semen harvesting with electroejaculation or vibratory stimulation is associated with a significant risk of autonomic dysreflexia.
    e. Pre-treatment of SCI men undergoing semen harvesting via vibratory stimulation with an α-adrenergic blocking drug such as tamsulosin minimizes the risk of severe hypertension.
15. Which of the following is true in men with retrograde ejaculation? (Select all that apply.)
    a. Retrograde ejaculation can be confirmed by the presence of spermatozoa in post-masturbation first-void urine.
    b. Retrograde ejaculation is more common following a bladder neck incision than transurethral resection of the prostate (TURP).
    c. Retrograde ejaculation may occur in men with diabetic autonomic neuropathy.
    d.Retrograde ejaculation in men with diabetic autonomic neuropathy is usually associated with hypogonadism.
    e. SSRIs are effective treatments for retrograde ejaculation.

Answers

1. **b. Serotonin.** Many neurotransmitters are involved in the control of ejaculation, including dopamine, norepinephrine, serotonin, acetylcholine, oxytocin, GABA, and nitric oxide (NO). Of the many studies conducted to investigate the role of the brain in the

development and mediation of sexual functioning, dopamine and serotonin have emerged as essential neurochemical factors. Whereas dopamine promotes seminal emission/ejaculation via D2 receptors, serotonin is inhibitory.

2. **e. Closure of bladder neck, rhythmic contraction of the bulbocavernosus, bulbospongiosus, and other pelvic floor muscles, and relaxation of the internal urinary sphincter during emission prevents retrograde ejaculation.** Based on functional, central, and peripheral mediation, the ejaculatory process is typically subdivided into three phases: emission, ejection (or penile expulsion), and orgasm. Emission consists of contractions of seminal vesicles (SV) and the prostate, with expulsion of sperm and seminal fluid into the posterior urethra, and is mediated by synergistic activation of both sympathetic and parasympathetic nerves (T10 to L2) but may also include a cholinergic excitatory mechanism. Ejection is mediated by somatic nerves (S2 to S4) within the pudendal nerve, and involves pulsatile contractions of the bulbocavernosus and pelvic floor muscles together with contraction of the internal urinary sphincter to prevent retrograde ejaculation. Ejection also involves a sympathetic spinal cord reflex upon which there is limited voluntary control. The bladder neck closes to prevent retrograde flow; the bulbocavernosus, bulbospongiosus, and other pelvic floor muscles contract rhythmically, and the internal urinary sphincter contracts. The external urinary sphincter exhibits intense contractions interrupted by silence periods during ejection of semen.
3. **b. 2.5%.** Community-based normative IELT research and observational studies of men with PE demonstrate that IELTs of less than 1 minute have a low prevalence of about 2.5% in the general population, although a substantially higher percentage of men with normal IELT complain of PE.
4. **b. Approximately 50% of heterosexual men seeking treatment for lifelong PE ejaculate within 1 minute after pen etration.** The medical literature contains several univariate and multivariate operational definitions of PE. Each of these definitions characterize men with PE using all or most of the accepted dimensions of this

condition: ejaculatory latency, perceived ability to control ejaculation, reduced sexual satisfaction, personal distress, partner distress, and interpersonal or relationship distress. The first official definition of PE was established in 1980 by the American Psychiatric Association (APA) in the DSM-III. The current APA DSM-V definition of PE is largely based upon the unified PE definition of the ISSM and captures the key dimensions of latency, control, and bother. Earlier DSM definitions did not include latency.

Several observational studies in cohorts of heterosexual men with lifelong PE with prospective stopwatch IELT measurement showed that about 90% of men seeking treatment for lifelong PE ejaculated within 1 minute after penetration, and about 10% ejaculated between 1 and 2 minutes. These data support the proposal that lifelong PE is characterized by an IELT of less than or about 1 minute after vaginal penetration.

Recent studies have suggested that in some men neurobiological and genetic variations could contribute to the pathophysiology of lifelong premature ejaculation (L-PE), as defined by the ISSM criteria and that the condition may be maintained and heightened by psychological/environmental factors. Acquired PE is commonly due to sexual performance anxiety, psychological or relationship problems, ED, and occasionally prostatitis, hyperthyroidism, or during withdrawal/detoxification from prescribed or recreational drugs.

Four PE subtypes are distinguished on the basis of the duration of the IELT, frequency of complaints, and course in life. In addition to lifelong PE and acquired PE, this classification includes natural variable PE (or variable PE) and premature-like ejaculatory dysfunction (or subjective PE). Men with subjective PE complain of PE, while actually having a normal or even extended ejaculation time. The complaint of PE in these men is probably related to psychological and/or cultural factors.

5. **e. This definition is applicable to all men regardless of their sexual orientation or type of sexual contact.** In October 2007, the ISSM convened an initial meeting of the first Ad Hoc ISSM

Committee for the Definition of Premature Ejaculation to develop the first contemporary, evidence-based definition of lifelong PE. Evidence-based definitions seek to limit errors of classification and thereby increase the likelihood that existing and newly developed therapeutic strategies are truly effective in carefully selected dysfunctional populations.

The committee unanimously agreed that the constructs that are necessary to define lifelong PE are time from penetration to ejaculation, inability to delay ejaculation, and negative personal consequences from PE, and they recommended the following definition. "*Lifelong PE is a male sexual dysfunction characterized by the presence of all of these criteria: (1) ejaculation that always or nearly always occurs prior to or within about 1 minute of vaginal penetration; (2) the inability to delay ejaculation on all or nearly all vaginal penetrations; and (3) negative personal consequences such as distress, bother, frustration, and/or the avoidance of sexual intimacy."*

In April 2013, a second Ad Hoc ISSM Committee for the Definition of Premature Ejaculation agreed that although lifelong and acquired PE are distinct and different demographic and etiological populations, they can be jointly defined, in part, by the constructs of time from penetration to ejaculation, inability to delay ejaculation, and negative personal consequences from PE. The committee determined that the presence of a clinically significant and bothersome reduction in latency time, often to about 3 minutes or less, was an additional key defining dimension of acquired PE. This definition is limited to men engaging in vaginal intercourse as there are few studies available on PE research in homosexual men or during other forms of sexual expression.

6. **d. Men with lifelong or acquired PE invariably experience a variety of negative psychological consequences such as bother, frustration, or the avoidance of sexual contact.** Waldinger et al. (1998) reported IELTs less than 30 seconds in 77% and less than 60 seconds in 90% of 110 men with lifelong PE with only 10% ejaculating between 1 and 2 minutes. These data are consistent with normative community IELT data, support the notion that IELTs of

less than 1 minute are statistically abnormal, and confirm that an IELT cut-off of 1 minute will capture 80% to 90% of treatment-seeking men with lifelong PE. A post hoc analysis of the dapoxetine Phase 3 COUPLE trial data confirms a statistically significant higher IELT in men with acquired PE and comorbid compared to men with lifelong PE with comorbid ED. Additional recent studies support this report and confirm that self-estimated IELT was lower in men with lifelong PE compared to acquired PE and highest in men with subjective PE. These data suggest 3 minutes as a valid cut-off for either self-estimated or stopwatch IELT for the diagnosis of acquired PE.

Several authors report that estimated and stopwatch IELT correlate reasonably well or are interchangeable in assigning PE status when estimated IELT is combined with patient-reported outcomes (PROs). Since patient self-report is the determining factor in treatment seeking and satisfaction, it is recommended that self-estimation by the patient and partner of ejaculatory latency is accepted as the method for determining IELT in clinical practice.

The ability to prolong sexual intercourse by delaying ejaculation and the subjective feelings of ejaculatory control comprise the complex construct of ejaculatory control. Virtually all men report using at least one cognitive or behavioral technique to prolong intercourse and delay ejaculation, with varying degrees of success, and many young men reported using multiple different techniques. Voluntary delay of ejaculation is most likely exerted either prior to or in the early stages of the emission phase of the reflex but progressively decreases until the point of ejaculatory inevitability.

Several authors have reported an association between lifelong or acquired PE and negative psychological outcomes in men with lifelong or acquired PE and their female partners. This personal distress has discriminative validity in diagnosing men with and without PE. The personal and/or interpersonal distress, bother, frustration, and annoyance that result from PE may affect men's quality of life and partner relationships, their self-esteem and self-confidence, and can act as an obstacle to single men forming new partner relationships.

7. **e. Subthyroidism is a common cause of PE.** Reliable information on the prevalence of lifelong and A-PE in the general male population is lacking. PE has been estimated to occur in 4% to 39% of men in the general community. Prevalence data derived from patient self-report is appreciably higher than prevalence estimates based on clinician diagnosis utilizing the more conservative ISSM definition of PE. As a result, there is a substantial disparity between the incidence of PE in epidemiological studies that rely upon either patient self-report of PE and/or inconsistent and poorly validated definitions of PE.

   Community-based stopwatch studies of the IELT, the time interval between penetration and ejaculation, demonstrate that the distribution of the IELT is positively skewed, with a median IELT of 5.4 minutes (range, 0.55 to 44.1 minutes), decreases with age and varies between countries, and supports the notion that IELTs of less than 1 minute are statistically abnormal compared to men in the general western population.

   Early ejaculation in humans has been explained by either hyposensitivity of the 5-HT2C and/or hypersensitivity of the 5-HT1A receptor. Recent studies have suggested that in some men neurobiological and genetic variations could contribute to the pathophysiology of L-PE, as defined by the ISSM criteria and that the condition may be maintained and heightened by psychological/environmental factors.

   The majority of patients with thyroid hormone disorders experience sexual dysfunction. Studies suggest a significant correlation between PE and suppressed thyroid-stimulating hormone (TSH) values in a selected population of andrological and sexological patients. One author reports that the 50% prevalence of PE in men with hyperthyroidism fell to 15% after treatment with thyroid hormone normalization. Although occult thyroid disease has been reported in the elderly hospitalized population, it is uncommon in the population who present for treatment of PE, and routine TSH screening is not indicated unless clinically indicated.
8. **b. PE is rarely compounded by the presence of high levels of performance anxiety related to their ED.** Recent data

demonstrates that as many as half of subjects with ED also experience PE. Subjects with ED may either require higher levels of stimulation to achieve an erection or intentionally "rush" intercourse to prevent early detumescence of a partial erection, resulting in ejaculation with a brief latency. This may be compounded by the presence of high levels of performance anxiety related to their ED, which serves to only worsen their prematurity.

ED is often associated with endothelial dysfunction and atherosclerosis of the internal pudendal and cavernous arteries due to the presence of vascular risk factors such as diabetes mellitus, hypertension, hyperlipidemia, and cigarette smoking.

Off-label on-demand or daily dosing of PDE-5 inhibitors is not recommended for the treatment of L-PE in men with normal erectile function. However, ED pharmacotherapy alone or in combination with PE pharmacotherapy is recommended for the treatment of L-PE or A-PE in men with co-morbid ED. Phosphodiesterase type-5 isoenzyme (PDE5) inhibitors, sildenafil, tadalafil, and vardenafil are effective treatments for ED. Several authors have reported experience with PDE-5 inhibitors alone or in combination with SSRIs as a treatment for PE.

9. **c. The index of premature ejaculation (IPE) was developed specifically for use as a screening questionnaire.** Men presenting with self-reported PE should be evaluated with a full medical/sexual history, a focused physical examination, inventory assessment of erectile function, and any investigations suggested by these findings. Current literature suggests that the diagnosis of L-PE is based purely on the medical history as there are no predictive physical findings or confirmatory investigations. However, in men with A-PE, a physical examination is mandatory in an effort to identify the etiology of the PE and to alleviate its possible cause. Laboratory or imaging investigations are occasionally required based upon the patient's medical history. A digital prostate examination, routine in an andrological setting for all men over 40, is useful in identifying possible evidence of prostatic inflammation or infection.

The presence of comorbid ED should be evaluated using a validated instrument such as the IIEF or the IIEF-5 (SHIM). The IIEF is not specifically validated in men with PE. Caution should be exercised in the IIEF diagnosis of comorbid ED in men with PE as 33.3% of potent men with PE confuse the ability to maintain erections prior to ejaculation and following ejaculation, record contradictory response/s to some/all questions of the SHIM especially Q3 and Q4, and receive a false positive IIEF/SHIM diagnosis of ED.

10. **c. Psychological-behavioral strategies for treating PE are moderately successful in the long term.** There are multiple psychosexual and pharmacological treatments for PE. Graded levels of patient and couple counseling, guidance, and/or relationship therapy, either alone or ideally in combination with PE pharmacotherapy, should be offered as a treatment option for most men with PE. All men seeking treatment for PE should receive basic psychosexual education or coaching.

Although the new and often more expedient pharmacological therapies are overshadowing traditional psychological-behavioral methods in the treatment of PE, the psychological-behavioral approach remains an attractive option but is time-consuming, requires substantial resources of both time and money, lacks immediacy, requires the partner's cooperation, and has mixed efficacy.

Psychological interventions are designed to achieve more than simply increasing the IELT. Targeted factors focus on the man, his partner, and their relationship. Psychotherapy and behavioral interventions improve ejaculatory control by helping men/couples to: (1) learn techniques to control and/or delay ejaculation, (2) gain confidence in their sexual performance, (3) lessen performance anxiety, (4) modify rigid sexual repertoires, (5) surmount barriers to intimacy, (6) resolve interpersonal issues that precipitate and maintain the dysfunction, and (7) increase communication.

Psychological-behavioral strategies for treating PE have been at least moderately successful in alleviating the dysfunction in the short term, but long-term outcome data are limited and suggest a significant relapse rate.

11. **c. SSRIs inhibit the postsynaptic 5-HT transporter system in the serotonergic neuron synapse.** Dapoxetine has received approval for the treatment of PE in over 50 countries worldwide. Dapoxetine has not received marketing approval for the United States by the Food and Drug Administration (FDA). It is a rapid-acting and short half-life SSRI with a pharmacokinetic profile supporting a role as an on-demand treatment for PE.

    Several forms of pharmacotherapy have been used in the treatment of PE. These include the use of topical local anesthetics (LAs), SSRIs, tramadol, phosphodiesterase type 5 inhibitors (PDE5i), and alpha adrenergic blockers. The use of topical LA, such as lidocaine, prilocaine, or benzocaine, alone or in association, to diminish the sensitivity of the glans penis is the oldest known pharmacological treatment for PE. The introduction of the SSRIs—paroxetine, sertraline, fluoxetine, citalopram, and the tricyclic antidepressant (TCA) clomipramine—has revolutionized the treatment of PE. These drugs block pre-synaptic axonal reuptake of serotonin from the synaptic cleft of central serotonergic neurons by 5-HT transporters, resulting in enhanced 5-HT neurotransmission and stimulation of post-synaptic membrane 5-HT receptors.

    On-demand administration of clomipramine, paroxetine, sertraline, and fluoxetine 3 to 6 hours before intercourse is modestly efficacious and well tolerated but is associated with substantially less ejaculatory delay than daily treatment in most studies.

12. **d. Many men with acquired DE can often masturbate to orgasm and may use idiosyncratic masturbation techniques that cannot be easily replicated during intercourse.** The DSM-IV-TR definition of delayed ejaculation contains no clear criteria as to when a man actually meets the conditions for DE, as operationalized criteria do not exist. As most sexually functional men ejaculate within about 4 to 10 minutes following intromission, a clinician might assume that **men with latencies beyond 25 or 30 minutes (21 to 23 minutes represents about 2 standard deviations above the mean) who report distress or men who simply cease sexual activity due to exhaustion or irritation qualify for this diagnosis.** Such symptoms, together with the fact

that a man and/or his partner decide to seek help for the problem, are usually sufficient for a DE diagnosis.

Psychogenic DE, often described as inhibited ejaculation, is usually related to sexual performance anxiety, which may draw the man's attention away from erotic cues that normally serve to enhance arousal. It is occasionally characterized by the use of idiosyncratic and vigorous masturbation styles that cannot be replicated during intercourse with a partner, or an "autosexual" orientation where men derive greater arousal and enjoyment from masturbation than from intercourse. These men precondition themselves to possible difficulty attaining orgasm with a partner and, as a result, experience acquired DE. These men appear able to achieve erections sufficient for intercourse despite a relative absence of subjective arousal, and their erections are taken as erroneous evidence by both the man and his partner that he was ready for sex and capable of achieving orgasm.

13. **b. Hypogonadism, d. Multiple sclerosis,** and **e. Antidepressant drugs such as SSRIs.** Delayed ejaculation/anejaculation is associated with several differing pathophysiologies that include congenital disorders as well as ones caused by psychological factors, treatment of male pelvic cancers with surgery or radiotherapy, neurological disease, endocrinopathy, infection, and treatment for other disorders. When a medical history or symptomatology so indicates, investigation of such possible etiologies may be necessary. The most common causes of DE seen in clinical practice are psychogenic inhibited ejaculation, degeneration of penile afferent nerves and Pacinian corpuscles in the ageing male, hypogonadism, diabetic autonomic neuropathy, treatment with SSRI antidepressants and major tranquillizers, radical prostatectomy, or other major pelvic surgery or radiotherapy.
14. **c. The ability to achieve an erection increases with descending levels of spinal injury** and **e. Pre-treatment of SCI men undergoing semen harvesting via vibratory stimulation with an α-adrenergic blocking drug such as tamsulosin minimizes the risk of severe hypertension.** The ability to ejaculate is severely

impaired by SCI. The level and completeness of SCI determine the post-SCI erectile and ejaculatory capacity. Unlike erectile capacity, the ability to ejaculate increases with descending levels of spinal injury. Less than 5% of patients with complete upper motor neuron lesions retain the ability to ejaculate. Ejaculation rates are higher (15%) in patients with both lower motor neuron lesions and an intact thoracolumbar sympathetic outflow. Approximately 22% of patients with an incomplete upper motor neuron lesion and almost all men with incomplete lower motor neuron lesions retain the ability to ejaculate. In those patients capable of successful ejaculation, the sensation of orgasm may be absent and retrograde ejaculation often occurs.

Several techniques for obtaining semen from SCI men with ejaculatory dysfunction have been reported. Vibratory stimulation is successful in obtaining semen in up to 70% of men with SCI. The use of electro-ejaculation to obtain semen by electrical stimulation of efferent sympathetic fibers of the hypogastric plexus is an effective and safe method of obtaining semen. Both vibratory stimulation and electro-ejaculation are associated with a significantly high risk of autonomic dysreflexia. Pre-treatment with a fast-acting vasodilator such as nifedipine minimizes the risk of severe hypertension, should autonomic dysreflexia occur with either form of treatment.

15. **a. Retrograde ejaculation can be confirmed by the presence of spermatozoa in post-masturbation first-void urine** and **c. Retrograde ejaculation may occur in men with diabetic autonomic neuropathy.** Antegrade (normal) ejaculation requires a closed bladder neck (and proximal urethra). Surgical procedures that compromise the bladder neck closure mechanism may result in retrograde ejaculation. The occurrence of orgasm in the absence of prograde ejaculation suggests retrograde ejaculation and can be confirmed by the presence of spermatozoa in post-masturbation first void urine.

Transurethral incision of the prostate (TUIP) results in retrograde ejaculation in 5% to 45% of patients and is probably related to whether one or two incisions are made and whether or not the

incision includes primarily the bladder neck or extends to the level of the verumontanum. The importance of contraction of the urethral smooth muscle at the level of the verumontanum has been hypothesized to be important in preventing retrograde ejaculation. TURP carries a higher incidence of retrograde ejaculation than does TUIP. The reported incidence of retrograde ejaculation following TURP ranges from 42% to 100%.

Retrograde ejaculation is more common in diabetes mellitus (DM) than in age-matched controls ($P < .01$), has been reported in 30% of men with DM, and is not statistically associated with duration of DM, body mass index (BMI), waist circumference, or HgbA1c or total testosterone levels.

Several sympathomimetic amine agents have been described as useful with mixed results. These drugs include pseudoephedrine, ephedrine, midodrine, and phenylpropanolamine. These agents work by stimulating the release of noradrenaline from the nerve axon terminals but may also directly stimulate both α- and β-adrenergic receptors. The TCA imipramine, which blocks the reuptake of noradrenaline by the axon from the synaptic cleft, is also occasionally useful. The usual dose is 25 mg twice daily. The current feeling is that long-term treatment with imipramine is likely to be more effective. While medical treatment may not always produce normal ejaculation, it may result in some prograde ejaculation.

Chapter Review

1. PE is a common sexual dysfunction.
2. PE is associated with negative psychological consequences including distress, bother, and frustration, which may affect quality of life, partner relationships, self-esteem, self-confidence, and can act as an obstacle to single men forming new partner relationships.
3. The evidence-based ISSM definition of lifelong and acquired PE should form the basis of the office diagnosis of lifelong PE.

4. There is limited evidence suggesting that lifelong PE has a genetic basis and acquired PE is most often due to sexual performance anxiety, psychological or relationship problems, and/or ED.
5. Oral SSRI drugs and topical anesthetic drugs are effective and safe treatments for PE.
6. Psychosexual cognitive behavioral therapy (CBT) has a limited role as a first-line treatment for PE but has an important role as an adjunct to pharmacotherapy, especially in men with acquired PE due to sexual performance anxiety.
7. Men with acquired PE most secondary to co-morbid ED, hyperthyroidism, chronic lower urogenital infection, prostatodynia, or chronic pelvic pain syndrome (CPPS) should receive appropriate etiology-specific treatment alone or in combination with an SSRI.
8. The causes of DE and anejaculation are manifold.
9. Failure of ejaculation can be a lifelong problem (25%) or an acquired problem (75%). It may be global and occur in every sexual encounter or be intermittent or situational.
10. Treatment of men with DE should be etiology-specific and address the issue of infertility in men of reproductive age.
11. Drug treatment of men with DE or anejaculation has limited success..
12. TURP and diabetic autonomic neuropathy are the most common causes of retrograde ejaculation.
13. RE and failure of emission can be distinguished by examination of a post masturbatory specimen of urine for the presence of spermatozoa and fructose.
14. Pharmacotherapy is associated with variable degrees of success and includes agents such as pseudoephedrine, midodrine, and imipramine.
15. The symptoms of post-orgasmic illness syndrome (POIS) occur following ejaculation and comprise severe myalgia, fatigue associated with a flu-like state, nasal congestion, and itching eyes.
16. POIS symptoms commence within 30 minutes of ejaculation in 87% of men and increase in intensity to a peak on day 2.
17. A type 1 hypersensitivity immunogenic reaction has been proposed as the underlying mechanism.

18. A prolonged hyposensitization program with multiple subcutaneous injections of autologous semen has been suggested as a possible treatment.

72

Surgery for Erectile Dysfunction

Matthew J. Mellon, and John J. Mulcahy

Keywords

treatment of erectile dysfunction; penile implant complications; penile implant insertion; penile implant; penile prosthesis

Questions

1. A 27-year-old patient presented to the emergency room with 36 hours of a prolonged painful erection. Blood gas analysis shows this to be veno-occlusive priapism, and bilateral T-shunts with corporal dilation were performed. This was unsuccessful in reducing the erection, and 1 week later the erection detumesced. After counseling, the patient elects to have a penile implant placed at this time. The type of cylinder that should NOT be used in this circumstance would be:
    a. AMS 700 CX.
    b. AMS 700 LGX.
    c. Genesis.
    d. Titan.
    e. Tactra.
2. During dilation to place a cylinder into a scarred corporal body, a dilator enters the urethra in the fossa navicularis. All of the following would be acceptable approaches EXCEPT:
    a. abandon the case.

b. proceed placing both cylinders.
c. place a cylinder on the side without the perforation and return at a later date to place the other cylinder.
d. create hypospadias, repair the perforation defect securely, and place both cylinders.

3. Select the TRUE statement:
   a. Repair of penile implants is currently in the range of 50% within 5 years.
   b. Satisfaction with penile implants among both patients and partners is in the 80% to 90% range.
   c. Infection rates with penile implant placement are comparable to those of other surgical procedures in the range of 5% to 7%.
   d. Semirigid rod prostheses are preferable to hydraulic implants in impotent spinal cord injury patients.
   e. The lock-out valves of the two vendors' three-piece inflatable penile implants designed to minimize autoinflation are in the same location.
4. A "salvage" or "rescue" procedure for an infected penile implant would likely be less successful in a patient with all of the following features EXCEPT:
   a. cellulitis around the incision and over prosthetic parts.
   b. purulent drainage from the incision.
   c. occurrence within 2 months of the implant placement.
   d. associated with an aggressive organism such as methicillin-resistant *Staphylococcus aureus* (MRSA) or *Pseudomonas aeruginosa.*
   e. diabetes mellitus as a comorbidity.
5. Which of the following is FALSE regarding informed consent for penile prosthesis surgery?
   a. The greatest disappointment after surgery is the size of the erection.
   b. Tissue problems such as pump migration, erosion, and extrusion of parts occur and require re-operation.
   c. Penile sensation and climax are not affected postoperatively.
   d. Alternative erectile dysfunction (ED) treatment options should be explained and offered prior to surgery.

e. Unrealistic expectations for the results of penile implant surgery should be identified and discussed prior to surgery.

6. Advantages to the penoscrotal approach for penile implant placement include all of the following EXCEPT:
   a. secure placement of the pump in a subdartos pouch.
   b. cosmetic skin incision.
   c. avoidance of damaging the dorsal neurovascular bundle.
   d. easier placement of semirigid rod implants.
   e. allows placement of the two-piece inflatable implant.
7. Select the FALSE statement:
   a. A high electrocautery setting may damage the Bioflex cylinders but not silicone-covered cylinders.
   b. Both the Bioflex and triple-layer AMS 700 cylinders are very durable, but may form aneurysms when subjected to repeated high external pressure.
   c. Semirigid rod implant malfunction is very rare and is limited to central wire breakage.
   d. The Ambicor cylinders give less rigidity and flaccidity than three-piece inflatable cylinders.
   e. The AMS 700 LGX cylinders are superior to the AMS 700 CX and Titan cylinders for straightening a curved erection due to their ability to increase in length.
8. The following are features of crural crossover of penile implant cylinders EXCEPT:
   a. difficult cylinder insertion.
   b. cylinders not appearing to be symmetric on either side of a Foley catheter in the urethra.
   c. curved erection.
   d. lopsided appearing erection.
   e. total corporal length measurements significantly different on right and left sides.
9. The following are suitable locations for a penile implant reservoir EXCEPT:
   a. a scrotal cavity where the testis had been previously removed.
   b. below the rectus muscle above the transversalis fascia.
   c. the epigastrium through a subcostal incision.

d. deep in the fat above the abdominal wall fascia in an obese patient.
e. in the peritoneal cavity.

Answers

1. **b. AMS 700 LGX.** The LGX is the only cylinder with distal expansion when inflated. Following a T-shunt, this cylinder may protrude through the defect at the end of the corporal body or create pressure on this weakened area when it is inflated.
2. **b. Proceed placing both cylinders.** Placing both cylinders would allow organisms to possibly enter the defect and contaminate the cylinder. In the other answers the cylinder is not exposed to possible infection.
3. **b. Satisfaction with penile implants among both patients and partners is in the 80% to 90% range.** Repair of penile implants is currently in the range of 15% at 5 years. Infection rates with penile implant placement are comparable to those of other surgical procedures in the range of 1% to 3%. Spinal cord injury patients have a high incidence of rod erosions (20% to 50%). Coloplast has the lock-out valve on the reservoir; BSci has the lock-out valve on the pump.
4. **e. Diabetes mellitus as a comorbidity.** Answers a through d are risk factors for failed salvage as noted in the chapter narrative. Diabetes mellitus, although a risk factor for implant infection, is not associated with a higher incidence of failed salvage.
5. **c. Penile sensation and climax are not affected postoperatively.** Penile sensation and ejaculation may both be diminished postoperatively.
6. **d. Easier placement of semirigid rod implants.** Semirigid rod cylinders are most easily placed through a subcoronal corporotomy.
7. **e. The AMS 700 LGX cylinders are superior to the AMS 700 CX and Titan cylinders for straightening a curved erection due to their ability to increase in length.** The LGX cylinders tend to exaggerate a curved erection.

8. **c. Curved erection.** Despite crossover, the erection is commonly straight.
9. **a. A scrotal cavity where the testis had been previously removed.** The Cloverleaf reservoirs contain lockout valves that could be manipulated through the scrotum, resulting in cylinder autoinflation. The Conceal reservoir is too large for scrotal placement, and the AMS spherical reservoir might fit with underfilling but would be subject to manipulation and wear in a scrotal location.

Chapter Review

1. The size of the erect penis after surgery will likely be shorter and thinner than preoperatively.
2. The implant provides only a firm penis suitable for intercourse.
3. Penile sensation and ejaculation may both be diminished postoperatively.
4. There will be varying degrees of postoperative pain.
5. Reoperation for a mechanical problem with the implant may be necessary.
6. Tissue problems such as infection may necessitate another surgery.
7. There are other treatments of ED such as pills, penile inserts and injections, and vacuum erection devices (VEDs).
8. There is a variety of implants including three-piece and two-piece inflatables and semirigid rods.
9. The three-piece inflatable implants give the best rigidity and flaccidity and are by far the most popular among both patients and urologists.
10. A shorter size to the erection following penile implant placement is the major reason for disappointment after surgery, and should be stressed in the informed consent.
11. Patients with limited dexterity should be encouraged to choose a semirigid rod implant to avoid frustration, when trying to manipulate the pump of a hydraulic device.

12. Injuries between the corpus cavernosum and the exterior should always be closed securely before an implant cylinder is left in place.
13. Placing a cylinder that does not expand in length will usually adequately straighten a curved erection in patients with tunica albuginea scaring, i.e., Peyronie disease.
14. Always keep the distal end of scissors or dilating instruments pointing dorsolaterally and parallel to the penile shaft to avoid crural crossover and urethral injury.
15. In the presence of a penile implant infection, all foreign material should be removed from the wound, as remnant parts may harbor microbes that perpetuate the infection.
16. In patients at risk for reduced penile vascular perfusion, be aware that distal necrosis can occur. Limit or stage aggressive reconstructive procedures in such patients.
17. Place the implant reservoir in a capacious cavity to avoid injuring surrounding structures such as bladder or bowel when the balloon expands as it is filled with isotonic saline.

7 3

Diagnosis and Management of Peyronie's Disease

Allen D. Seftel, and Hailiu Yang

Keywords

Clostridium histolyticum; collagenase; erectile dysfunction; interferon; male hypogonadism; Nesbit procedure; penile graft; penile modeling; penile prosthesis; Peyronie's disease; vacuum erection device

Questions

1. Peyronie's disease (PD):
    a. is a wound healing disorder.
    b. is an autoimmune disease.
    c. is a transient disorder (30% to 50% of the time).
    d. may degenerate into cancer.
    e. is associated with Dupuytren contracture in 40% of men.
2. The fibrous plaques in PD originate in:
    a. Buck's fascia.
    b. corpora cavernosa.
    c. corpus spongiosum.
    d. the tunica albuginea surrounding the corpora cavernosa.
    e. the tunica albuginea surrounding the corpus spongiosum.
3. Based on recent natural history studies on PD, a 54-year-old man with a 60-degree dorsal curve 18 months after onset has:

a. 40% chance of spontaneous improvement of deformity.
b. 70% chance of getting worse.
c. 10% chance of getting worse.
d. 10% or less chance of spontaneous improvement of deformity.
e. 80% chance of staying the same.

4. What is the prevalence of PD following radical prostatectomy?
   a. 4%
   b. 1% to 3%
   c. 11% to 16%
   d. >20%
   e. 0%
5. Plaque calcification:
   a. can be reliably identified on physical examination.
   b. can be found in 50% to 60% of men with PD.
   c. is not an indication of stable, mature disease.
   d. is associated with successful intralesional injection therapy.
   e. is a predictive factor for the need for surgical treatment.
6. Psychological distress in men with PD:
   a. is infrequent.
   b. is typically resolved by successful surgery.
   c. is frequently associated with penile shortening.
   d. correlates with degree of erect curvature.
   e. has no association with relationship issues with the patient's partner.
7. All of the following statements regarding the physical examination of a man presenting with PD are true EXCEPT that the examination:
   a. should include measurement of plaque size with calipers.
   b. should include direct assessment of curvature following injection of vasoactive agent.
   c. should include measurement of stretched penile length.
   d. can be supported with a picture of the erect penis by a smartphone.
   e. should include assessment of the patient's palms.
8. Penile duplex ultrasound provides information on all of the parameters below EXCEPT:

a. penile deformity.
b. erectile response to vasoactive penile injection.
c. penile vascular flow parameters.
d. penile sensory integrity.
e. plaque calcification.

9. Pentoxifylline and the phosphodiesterase-5 (PDE5) inhibitors have been shown in an animal model of PD to reduce scarring by what proposed mechanism?
   a. Improved penile blood flow
   b. Antiinflammatory
   c. Elevated local levels of nitric oxide (NO)
   d. Anti–transforming growth factor-β (TGF-β)
   e. Mechanotransduction
10. Penile injection therapy (i.e., prostaglandin E1, TriMix-gel, etc.) for ED is not directly responsible for:
   a. cavernosal fibrosis.
   b. Peyronie's disease.
   c. penile pain.
   d. priapism.
   e. high success rate.
11. A 35-year-old man with PD is able to achieve an erection adequate for intercourse with minimal discomfort and reported dorsal erect penile curvature of 20 degrees. The initial treatment should be:
   a. reassurance.
   b. oral vitamin E.
   c. intralesional steroids.
   d. oral tamoxifen.
   e. intralesional collagenase.
12. A 66-year-old man presents with a 2-year history of PD and a 55-degree dorsal curvature. He also notes that his average-grade erection with sexual stimulation at home is a grade {5/10}, which would not be adequate for intromission even if the penis were straight. Duplex ultrasound analysis demonstrates arterial insufficiency and an inadequate erectile response to 90 mg of intracorporal papaverine. The most appropriate treatment option for this patient who wants to resume sexual activity would be:

a. 16-dot plication.
b. vacuum constriction device.
c. penile prosthesis with penile straightening.
d. intralesional verapamil injections.
e. Plaque incision and grafting

13. Tunica plication is preferred for mild to moderate lateral curvature correction in a patient who is able to accomplish adequate erections with the assistance of PDE-5I because of:
    a. less shortening compared with grafting.
    b. better sensory protection.
    c. diminished risk of postoperative ED.
    d. greater potential for loss of erect length.
    e. less chance of recurrence
14. Postoperative rehabilitation is designed to aid in postoperative healing and outcomes in the following ways, EXCEPT:
    a. to prevent shortening and possibly recover some lost length.
    b. to encourage straight healing.
    c. to enhance cicatrix contracture.
    d. to preserve vascular integrity.
    e. to encourage partner participation.
15. All of the following are true regarding penile traction therapy after surgical penile straightening, EXCEPT:
    a. increases or preserves postoperative length.
    b. encourages tissue remodeling.
    c. should be used for 3 or more hours/day for optimum results.
    d. increases the risk of sensory change.
    e. results appear dose-related.
16. What is the most dreaded complication occurring after manual modeling during placement of a penile prosthesis in a man with PD?
    a. Tunica tear proximal
    b. Urethral injury
    c. Sensory deficit
    d. Recurrent curvature
    e. Distal urethral perforation

17. Indications to perform a plaque incision or partial excision and grafting include all of the following, EXCEPT:
    a. severe curvature in excess of 70 degrees.
    b. indentation resulting in an unstable penis or hinge effect.
    c. a short penis (<9 cm) with severe curve and poor rigidity.
    d. a short penis with severe curvature and excellent rigidity.
    e. extensive plaque calcification associated with severe deformity.
18. The primary concern with a grafting procedure for PD is:
    a. postoperative erectile dysfunction (ED).
    b. incomplete correction of hinge effect.
    c. penile shortening.
    d. decreased penile sensation
    e. graft infection.
19. Which of the following is a candidate for plaque incision and grafting?
    a. 60-degree severe ventral curve
    b. 30-degree curvature with suboptimal rigidity even with PDE5 inhibitors
    c. 70 degrees with hinge effect
    d. 90-degree lateral curve presenting 5 months after onset
    e. 30-degree dorsal curvature in a patient who desires to gain length as a result of surgery
20. The ideal graft includes all of the following EXCEPT:
    a. thin, strong, and easy to suture.
    b. no rejection.
    c. resistant to infection.
    d. preserves erectile function.
    e. contracts during healing.
21. The IMPRESS trials for Clostridium Collagenase Histolyticum (CCH) with modeling found that which of the following patient group would benefit from CCH?
    a. Calcified plaques that have been present for less than 12 months
    b. Ventral plaques greater than 60 degrees
    c. Stable plaques between 30 and 90 degrees.

d. Stable ventral plaques less than 60 degrees
e. Hourglass deformities less than 45 degrees.

22. Which of the following is a significant concern with CCH?
    a. Cost and length of treatment
    b. Risk of ED
    c. Autoimmune reaction
    d. Penile bruising
    e. High risk of penile fracture

Answers

1. **a. Is a wound healing disorder. PD is currently recognized as a wound healing disorder of the tunica albuginea (Devine and Horton, 1988)** [a] that results in the formation of an exuberant scar, occurring presumably after an injury to the penis, which activates an abnormal wound healing response (Ralph et al., 2010; Levine and Burnett 2013; Greenfield and Levine, 2005; Van De Water, 1997). PD is not a premalignant condition, spontaneous resolution is a rare event, and improvement likely does not take place in more than 13% of men over the first 12 to 18 months. Though an association with Dupuytren contracture has been described, studies vary widely on this association.
2. **d. The tunica albuginea surrounding the corpora cavernosa.** PD plaques originate in the tunica albuginea. **Sixty percent to seventy percent of plaques are located on the dorsal aspect of the tunica albuginea and are usually associated with the septum (Pryor and Ralph, 2002).** It is possible that pressures on the penis during intercourse result in a delamination between the two layers, activating the abnormal wound-healing process, which is trapped within the tunic, fostering the progressive scarring.
3. **d. 10% or less chance of spontaneous improvement of deformity. Spontaneous regression has been looked at in several contemporary natural history studies, which have suggested that no more than 13% will experience improvement of deformity.** Full spontaneous resolution is extremely rare. If no

treatment is offered, up to 50% will experience worsening of their deformity (Mulhall et al., 2006).

4. **c. 11% to 16%.** Tal and associates (2010) demonstrated an incidence of PD of 15.9% with a mean time to development of disease of 13.9 months. Ciancio and Kim (2000) also examined the effects of prostatectomy on penile fibrosis and sexual dysfunction. **Eleven percent of all patients undergoing prostatectomy developed fibrotic changes in the penis.** This fibrosis led to penile curvature in 93%, "waistband" deformity in 24%, and palpable plaques in 69%.
5. **c. Is not an indication of stable, mature disease. Only recently has it been recognized that calcification may occur early after the onset of the scarring process, and therefore the previously held notion that calcification is an indication of chronic, severe, and/or mature disease appears untrue (Levine et al., 2013).** Several investigators have indicated that intralesional injection therapy of verapamil and interferon is less likely to be successful in men with significant calcification (Levine et al., 2002b; Hellstrom et al., 2006). A "rock-hard" plaque may be an indicator of calcification but will need to be confirmed with some form of imaging, preferably ultrasound. A calcified plaque is readily identified by ultrasound because of the hyperdensity of the plaque with shadowing behind it. Calcification itself does not predict the need for surgery. Approximately 34% of PD patients will have some degree of plaque calcification.
6. **c. Is frequently associated with penile shortening. Penile shortening and inability to have intercourse are the two most common and consistent risk factors for emotional distress and relationship problems associated with PD (Smith et al., 2008; Rosen et al., 2008).** Psychosocial stress is common and is reported by 77% to 94% of men with PD (Gelbard et al., 1990; Tal et al., 2012; Nelson and Mulhall, 2013). PD also commonly affects the patient's sexual partner, causing feelings of helplessness, as well as feeling personally responsible for the PD due to trauma during intercourse and sadness over loss of intimacy (Rosen et al., 2008). For some patients even a lesser degree of curvature may be highly

bothersome or provoke distress (Hellstrom et al., 2013). Despite "successful treatment" that may allow the patient to be sexually functional again, there is often persistent psychological distress, presumably due to the residual changes compared with that man's pre-PD penis (Jones, 1997; Gelbard et al., 1990).

7. **a. Should include measurement of plaque size with calipers.** Measurement of the size of the plaque with any modality has been found to be inaccurate, as the plaque is rarely a discrete lesion (Bacal et al., 2009; Levine and Burnett, 2013; Ralph et al., 2010; Hatzimouratidis et al., 2012). Deformity assessment via ultrasound after injection of vasoactive agent has been shown to be the best method of assessing curvature as well as erectile response. Pictures taken from multiple vantage points may give a better idea of deformity during initial consultation. Because of the association with other collagen vascular disorders, the patient's palms should be examined.
8. **d. Penile sensory integrity.** The benefits of a complete duplex ultrasound assessment include identification of calcification during initial surveillance in the flaccid state, assessment of penile vascular flow parameters following intracavernosal injection of vasoactive agent, observing the erectile response to the vasoactive injection compared to the patient's sexually induced erection at home, and finally providing the best opportunity to objectively assess deformity. These parameters are absolutely critical to the decision process for the patient who is considering surgery. Penile sensation is best evaluated with biothesiometry.
9. **c. Elevated local levels of nitric oxide (NO). Pentoxifylline is a nonspecific phosphodiesterase inhibitor with combined antiinflammatory and antifibrogenic properties.** NO synthesized by inducible nitric oxide synthase (iNOS) reacts with reactive oxygen species (ROS), thus reducing ROS levels and presumably inhibiting fibrosis. The antifibrotic effects of NO may be mediated at least in part by the reduction of myofibroblast abundance and lead to a reduction in collagen I synthesis (Vernet et al., 2002).
10. **b. Peyronie's disease.** Injection therapy has no association with leading to PD. It may, however, lead to some degree of cavernosal

fibrosis, pain, and priapism. When used in the appropriate population, intracavernosal injection therapy for ED does have a high success rate.

11. **a. Reassurance.** In this case, a young man presents with PD with a minimal curvature and minimal discomfort. Pursuing aggressive therapy is not indicated because the disease process may not worsen. Therefore, reassurance is the proper answer. This patient should also be counseled to follow up should he see any exacerbation of his symptoms. At this time, intralesional collagenase is not indicated for curvature less than 30 degrees, the two noted oral therapies are not noted to be beneficial, and intralesional steroids are not recommended because of lack of objective evidence of benefit.
12. **c. Penile prosthesis with penile straightening.** This man presents with stable PD, a borderline moderate to severe curvature, but a poor-quality erection at home and an inadequate erectile response during duplex ultrasound with a high dose of vasoactive intracorporal drug injection. For the motivated patient who has both ED and PD, placement of a penile prosthesis with straightening maneuvers is the most likely treatment to address both problems. Plication or plaque incision with grafting would not fix this man's ED. Use of oral therapy had not been proven to be beneficial in this circumstance. Intralesional injection of any sort may potentially benefit his deformity, but given that he has inadequate erectile response, intralesional injections would not result in ability to resume sexual activity.
13. **c. Diminished risk of postoperative ED.** A tunica plication procedure is the preferred approach because it has the least likelihood of causing injury to the underlying cavernosal tissue, which presumably is responsible for postsurgical ED seen most commonly with a grafting procedure. There is indeed a greater potential for loss of erect length with a plication and less likelihood of damaging the penile sensory nerves, but the diminished risk of postoperative ED is the strongest reason to pursue a plication rather than grafting procedure, particularly in patients with less than severe curvature.

14. **c. To enhance cicatrix contracture.** Postoperative rehabilitation with massage and stretch, nightly use of a PDE5 inhibitor, and traction therapy are designed to enhance postoperative healing in all listed ways except for c, which is the correct answer because enhancing cicatrix contracture would not enhance healing or postoperative outcomes.
15. **d. Increases the risk of sensory change. Penile traction therapy has emerged as an effective adjunct treatment option.** Traction therapy has been shown to increase or preserve postoperative length and encourage tissue modeling, and it does appear to be dose-related. In addition, one study suggested that the minimum average time for daily use following surgery would be 3 hours or more per day. There is no evidence that traction therapy causes injury to the sensory nerves, nor is there evidence to suggest that it increases the risk of sensory change.
16. **b. Urethral injury.** Manual modeling was introduced in the mid-1990s as a straightening maneuver to correct residual curvature after placement of a penile prosthesis. In performing this procedure, the primary reported risk is injury to the urethra at the meatus, where the prosthetic cylinder tips may extrude through the meatus as a result of pressure placed on the distal shaft during the modeling process. Although sensory deficit, recurring curvature, and tunica tear are possible complications, they have not been reported, nor has a proximal urethral perforation.
17. **c. A short penis (<9 cm) with severe curve and poor rigidity.** This is the only option noted that would not be appropriately treated by a grafting procedure for PD, primarily because of the poor rigidity. **One of the primary indications to perform a grafting procedure would be the patient having a strong erection preoperatively with severe curvature in excess of 70 degrees, having indentation causing an unstable penis or hinge effect, hourglass deformity, or a short penis. The key here is the poor rigidity, which would be an absolute contraindication to performing a grafting procedure; it is more likely to cause further ED in this circumstance.**

18. **a. Postoperative erectile dysfunction (ED).** Although the other listed side effects have been reported, the primary concern with patients undergoing this procedure is postoperative ED. Therefore, proper patient selection preoperatively would include only those men who have excellent-quality rigidity with or without PDE5 inhibitor therapy and have a normal vascular response during duplex ultrasound assessment.
19. **c. 70 degrees with hinge effect. Severe curve greater than 70 degrees and hinge effect are two indications to perform an incision and grafting procedure. Ventral curvatures when repaired with grafting have a much higher rate of complete ED.** Those with suboptimal rigidity tend to also develop more erectile problems post-grafting. A 90-degree curve may be an indication, but all surgical procedures should be delayed at least 1 year from the time of onset with 6 months of stable disease. Finally, it should be recognized that although a patient is more likely to recover some length after performing a grafting procedure, neither grafting nor plication operations should be expected to result in substantial gain of length; the primary goal is straightening.
20. **e. Contracts during healing.** Contraction during the healing process would not be included in the criteria for an ideal graft.
21. **c. Stable plaques between 30 and 90 degrees.** IMPRESS Phase 3 Randomized clinical trials (RCTs) showed that CCH resulted in an improvement of 17 degrees versus 9 degrees (control) in patients with stable dorsal plaques between 30 and 90 degrees. Those who had unstable hinges, calcified plaques, hourglass effect, severe ED, or ventral plaques were excluded from the study. Various quality of life parameters were also improved with CCH injection.
22. **a. Cost and length of treatment.** In the IMPRESS trial, patients were given four treatment courses consisting of two injections each. Each injection costs $3300. This leads to significant cost and numerous office visits, which may pose a significant burden to the patient. De novo ED has not been reported with CCH. Penile bruising is a common side effect but is almost always benign. Penile fracture is a dreaded complication but occurs in less than 1%

in a pooled analysis. There are no reports of severe autoimmune reactions associated with CCH.

Chapter Review

1. There are two phases in the natural history of PD: the acute phase, in which changes occur, and the stable chronic phase, which is typically marked by 12 months onset and 6 months of stable curvature.
2. The incidence of PD is between 3% and 9% with a peak age of 50 years.
3. There is a high association (33%) of diabetes with PD.
4. Preoperative erectile function correlates strongly with postoperative results.
5. Patients with poor-quality erections preoperatively who have grafting procedures are likely to have significant problems postoperatively with erectile function. These patients should be steered toward inflatable penile prosthesis (IPP) placement with or without adjunct procedures.
6. PD is currently recognized as a wound healing disorder of the tunica albuginea.
7. Sixty percent to seventy percent of plaques are located on the dorsal aspect of the tunica albuginea and are usually associated with the septum.
8. The natural history of PD studies suggest that no more than 13% of patients will experience improvement of deformity.
9. Eleven percent of all patients undergoing prostatectomy developed fibrotic changes in their penis.
10. Penile shortening and inability to have intercourse are the two most common and consistent risk factors for emotional distress and relationship problems associated with PD.
11. Primary indications to perform a grafting procedure in a patient with a strong erection are severe curvature in excess of 70 degrees, an indentation causing an unstable penis or hinge effect, and a short

penis. Poor rigidity is an absolute contraindication to performing a grafting procedure.

12. Ventral curvatures, when repaired with grafting, have a much higher rate of complete ED.
13. IMPRESS Phase 3 RCTs showed that CCH resulted in an improvement of 17 degrees versus 9 degrees (control) in patients with stable dorsal plaques between 30 and 90 degrees. Those who had unstable hinges, calcified plaques, hourglass defects, severe ED, or ventral plaques were excluded from the study. However, cost of treatment presents a significant barrier.
14. Injection site pain and bruising are common and usually benign. However, one should always suspect the rare but dreaded complication of penile fracture in these patients.
15. While graft incision/partial excision do typically result in less penile length loss versus plication, a patient should not expect to increase his penile length with any procedure.

---

[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, ed 12* , on the Expert Consult website.

74

Sexual Function and Dysfunction in the Female

Ervin Kocjancic, Valerio Iacovelli, and ömer Acar

Keywords

biopsychosocial assessment; disability; dysfunction; female sexual response; lower urinary tract dysfunction; transgender; treatment

Questions

1. Sexual health encompasses which of the following concepts?
    a. Absence of sexual dysfunction/problem
    b. Mental well-being
    c. Human development and maturation
    d. All of the above
    e. a and c
2. Which of these molecules is thought to play only a minor role in female genital sexual response?
    a. Vasoactive intestinal polypeptide
    b. Nitric oxide
    c. Acetylcholine
    d. Norepinephrine
    e. Aquaporins
3. The Female Sexual Function Index (FSFI) assesses all but which of the following aspects of sexuality?
    a. Sexual distress

b. Sexual desire
c. Sexual arousal
d. Sexual pain
e. Orgasm

4. Which of the following is NOT an essential part of the physical examination in a woman with sexual concerns?
    a. Vital signs
    b. Assessment of vaginal pH
    c. Palpation of the levator ani musculature
    d. Careful inspection of the vulva
    e. Biothesiometry
5. Which of the following statements is TRUE?
    a. Assessment of the patient's intimate relationship(s) is a key aspect of treating sexual problems.
    b. Women with spinal cord injury cannot experience orgasm.
    c. A linear pattern for sexual response is typical for all women.
    d. Survey instruments may take the place of history in evaluation of sexual concerns.
    e. All women who have sex with women identify as lesbian or bisexual.
6. Which of the following statements is FALSE?
    a. Hysterectomy may improve or worsen sexual function in women depending on the indication.
    b. Low serum testosterone levels have been clearly linked to worse sexual function in all women.
    c. Phosphodiesterase type 5 inhibitors are not currently approved for the management of problems with sexual arousal response in women.
    d. Sexual activity during routine pregnancy is safe.
    e. Women may have genital arousal responses to erotic materials that they find mentally or emotionally unappealing.
7. Which of the following are potential adverse events associated with supplemental testosterone in women?
    a. Hirsutism
    b. Acne
    c. Decreased high-density lipoprotein

d. Vaginal bleeding
e. All of the above

8. Decreased sexual interest/desire has been associated with which of the following conditions in women?
    a. Use of antidepressants
    b. Life stressors
    c. Hypoestrogenism
    d. Relationship problems
    e. All of the above
9. Education on sexuality is always indicated; which of the following women is likely to also benefit from medical and/or psychological treatment?
    a. 24-year-old woman who does not climax with vaginal penetration but does climax with clitoral stimulation
    b. 56-year-old woman with vaginal dryness that is well managed with sexual lubricant
    c. 35-year-old woman with bothersome decline in sexual desire
    d. All of the above
    e. None of the above
10. Which of the following diagnoses are included in the *Diagnostic and Statistical Manual of Mental Illness Fifth Edition (DSM-5)*?
    a. Female orgasmic disorder (FOD)
    b. Hypoactive sexual desire disorder
    c. Genitopelvic pain/penetration disorder
    d. a and c above
    e. All of the above
11. Which of the following conditions has been associated with lower bioavailable androgen levels in women?
    a. Hormonal contraceptives
    b. Surgical menopause
    c. Elevated prolactin levels
    d. All of the above
    e. None of the above
12. Which of the following has NOT been shown to be useful in the management of problems with sexual arousal in women?
    a. Topical prostaglandins

b. Muscle relaxants
c. Vaginal lubricants
d. Hormonal supplementation with androgens and/or estrogens
e. Psychosocial counseling

13. Which of the following has been definitively linked to sexual dysfunction in women?
    a. Postmenopausal estrogen replacement
    b. Obesity
    c. High educational achievement
    d. Depression
    e. Metabolic syndrome
14. Which of the following has the least evidence for efficacy in management of antidepressant-associated sexual dysfunction in women?
    a. Use of an adjunctive antidepressant
    b. Reassurance
    c. Drug cessation
    d. Drug holiday
    e. Sildenafil
15. What is the most commonly purported etiology for the sexual problems that occur in some women using hormonal contraception?
    a. Reduction of bioavailable testosterone
    b. Reduction of bioavailable estrogen
    c. Psychological distress
    d. Partner dissatisfaction
    e. Alteration of vascular response
16. Which of the following is the possible anatomic location of the Grafenberg spot (G-spot), which is particularly sensitive to tactile stimulation in some women?
    a. Anterior wall of the vagina at the level of midurethra
    b. Cervical canal
    c. Clitoral surface
    d. Posterior vaginal wall
    e. Perianal ring

17. Choose the correct statement regarding the laboratory evaluation for female sexual dysfunction.
    a. It is mandatory in all cases
    b. Normal range of testosterone levels have been established in women
    c. It is rarely indicated unless there is suspicion for an underlying contributory medical condition
    d. Thyroid dysfunction is the most commonly encountered hormonal abnormality in female patients with sexual dysfunction
    e. Abnormalities in hormone levels provide extensive diagnostic and prognostic utility in female sexual dysfunction
18. Choose the correct statement regarding FSFI.
    a. Contains 29 items
    b. Has an abbreviated version that can be used for screening pre- and postmenopausal women
    c. It is not validated for use in patients with cancer
    d. Maximum total score is 30
    e. It is not validated for use in research
19. Which of the following is the most widely studied, most validated physiological test used in the assessment of female sexual function?
    a. Laser Doppler imaging
    b. Magnetic resonance imaging
    c. Thermistor clip
    d. Vaginal photoplethysmography
    e. Doppler ultrasonography
20. Choose the correct statement regarding the DSM-5 definitions of sexual dysfunctions in women:
    a. All disorders, including genitopelvic pain/penetration disorder, require that the symptoms meet the DSM-5 definitions of that condition.
    b. Symptoms should be present for 3 months
    c. Symptoms should occur on at least half of the sexual occasions
    d. Symptoms should cause clinically significant distress

e. Symptoms can be a consequence of a severe relationship distress

21. Which of the following is the most prevalent type of female sexual dysfunction?
    a. Hypoactive sexual desire disorder
    b. Arousal disorder
    c. Orgasmic disorder
    d. Genitopelvic pain/penetration disorder
    e. Vaginismus
22. Choose the correct statement about the mechanism of action of Flibanserin, which is currently the only FDA-approved treatment for acquired, generalized hypoactive sexual desire disorder.
    a. Phosphodiesterase type 5 inhibitor
    b. α-adrenergic antagonist
    c. 5-HT1A receptor agonist and 5-HT 2A receptor antagonist
    d. 5-HT1A receptor partial agonist
    e. Estrogen receptor modulator
23. Choose the correct statement regarding FOD:
    a. Orgasmic disorder almost always occurs as an isolated form of female sexual dysfunction
    b. Selective serotonin reuptake inhibitors (SSRIs) are used to treat FOD
    c. The information gathered by the questionnaires about female sexual function is sufficient to arrive at the diagnosis of FOD
    d. DSM-5 definition of FOD does not contain the criterion that requires orgasmic difficulty to occur despite a normal excitement phase
    e. Systemic testosterone treatment (either alone or in combination with systemic/local estrogen treatment) has been approved by the FDA for the treatment for FOD
24. Which of the following is one of the criteria that have been suggested by Leiblum and Nathan and used in the diagnosis of persistent genital arousal disorder in women?
    a. Physiologic sexual arousal can be brief and remit on its own
    b. Symptoms of sexual arousal can be triggered by no stimulus at all

c. Symptoms of sexual arousal are not distressing
d. The signs of physiologic sexual arousal remit with ordinary orgasmic experience
e. The signs of physiologic sexual arousal are experienced in the presence of subjective feelings of sexual desire and arousal

25. Choose the correct statement regarding provoked vestibulodynia (PVD):
    a. Patients with primary PVD appear to have greater nerve fiber density and thicker vulvar vestibule compared to the secondary group
    b. Pelvic floor muscles are usually hypotonic in both primary and secondary PVD
    c. Positive family history and an association with enuresis is more commonly encountered in secondary PVD
    d. Primary PVD is more common than its secondary counterpart
    e. Local signs of inflammation are a common part of the clinical picture in primary vestibulodynia

Answers

1. **d. All of the above.** Sexual wellness incorporates many aspects of human experience.
2. **c. Acetylcholine.** Acetylcholine is thought to play a relatively minor role in sexual response in women.
3. **a. Sexual distress.** The FSFI does not include a metric to quantify or measure sexuality-related distress.
4. **e. Biothesiometry.** Biothesiometry may be indicated in some cases of genital neuropathy, but much of the same information can be gleaned from careful history and physical examination (with or without basic sensory testing).
5. **a. Assessment of the patient's intimate relationship(s) is a key aspect of treating sexual problems.** Women with spinal cord injury may experience orgasm, and women may endorse a circular or linear sexual response. A history is critical to evaluation of

sexual wellness, and up to half of women who have had sex with another woman do not identify as lesbian or bisexual.

6. **b. Low serum testosterone levels have been clearly linked to worse sexual function in all women.** There are data to support a role for testosterone in sexual function in some women, but this is the least well supported of the statements in this question.
7. **e. All of the above. These are well-established potential effects of testosterone. There are substantial concerns about the potential for cardiovascular disease or neoplasia, but robust data on risk are scant.**
8. **e. All of the above.** There are numerous potential causes of low sexual desire.
9. **c. 35-year-old woman with bothersome decline in sexual desire.** Many sexually healthy women do not climax with vaginal penetration. Use of vaginal lubricant is an effective and safe management option for vaginal dryness.
10. **d. a and c above.** The DSM V combined hypoactive sexual desire disorder and female sexual arousal disorder into female sexual interest/arousal disorder. Similarly, dyspareunia and vaginismus were combined into genitopelvic pain/penetration disorder. FOD was carried over from DSM IV TR.
11. **d. All of the above. Hormonal contraception, prolactin, and surgical menopause all tend to decrease bioavailable testosterone.**
12. **b. Muscle relaxants.** Muscle relaxants have been used with good efficacy for issues of sexual pain but have not been shown to directly aid sexual arousal response in women.
13. **d. Depression.** Depression is unambiguously linked to sexual dysfunction in women; the other entities have been linked to sexual dysfunction in some but not all studies.
14. **b. Reassurance.** There are peer-reviewed, published data to support all but answer b. Reassurance may be indicated for some women but may be viewed by others as a dismissal of their concerns related to antidepressant treatment.
15. **a. Reduction of bioavailable testosterone.** Hormonal contraception has been clearly linked to lower bioavailable serum

testosterone. Other etiologies may contribute.

16. **a. Anterior wall of the vagina at the level of midurethra.** A region of the anterior wall of the vagina overlying the midurethra has been identified as the Grafenberg spot (or G-spot), an area that, in some women, is particularly sensitive to tactile stimulation. Grafenberg's original article identified the erogenous area as arising from the urethra. Controversy still exists over the possible anatomic structure and site of the G-spot.
17. **c. It is rarely indicated unless there is suspicion for an underlying contributory medical condition.** Laboratory evaluation for female sexual dysfunction (FSD) is rarely indicated unless there is a suspicion for a specific medical condition that might explain patient's sexual problems. Therefore, random testosterone measurement is worthless in the evaluation of FSD. Furthermore, there are no data to justify serum testosterone measurement for FSD unless there is a concern about hyper/hypoandrogenic state. Normal range of testosterone levels have not been established in women, and testosterone levels do not correlate with libido in females. Likewise, progesterone, measured either alone or in combination with estrogen, has very limited diagnostic/prognostic utility in FSD. If a hormonal abnormality is suspected based on the systemic evaluation of the patient, then a screening blood workup can be done to rule out prolactinoma, thyroid dysfunction, and adrenal disorders. If adrenal insufficiency is suspected, measurement of dehydroepiandrosterone sulfate (DHEA-S) is useful because of isolated adrenal production.
18. **b. Has an abbreviated version that can be used for screening pre- and postmenopausal women.** FSFI is a brief, 19-item, multidimensional self-report instrument that assesses six domains of sexual function: desire, arousal, lubrication, orgasm, satisfaction, and pain. Target population consists of hetero- and homosexual women. It is available for pre- and postmenopausal women and women with medical and sexual disorders. It refers to the past 4 weeks and administration time varies between 10 and 15 minutes. Each domain of FSFI is scored on a scale of 0 to 6 points with the exception of the desire domain (scored from 1.2 to 6) and the

satisfaction domain (scored from 0.8 to 6). The instrument thus has a range of 2 to 36, with 36 representing a "perfect" score. The maximum total score is 36 and higher scores indicate better sexual functioning. A total FSFI score of less than 26.55 is considered at risk for sexual dysfunction. Although it is validated for use in research, it is not yet used in clinical practice. An abbreviated version (six items) also has been validated in pre- and postmenopausal women and has been proposed as a tool for screening women likely to have FSD. The FSFI also has been validated for use in patients with cancer, in whom strong psychometric properties have been reported in addition to women with chronic pelvic pain.

19. **d. Vaginal photoplethysmography.** There are many physiological monitoring parameters of sexual arousal, which could potentially assist in the diagnosis of organic diseases contributing to sexual dysfunctions. Genital blood flow can be measured with vaginal photoplethysmography, which is the most widely studied and most validated physiological test used in assessment of female sexual function. This method uses a vaginal light source on an acrylic tampon to illuminate the vaginal microcirculation and determine the level of vaginal engorgement. Most studies comparing genital responses of women with and without sexual dysfunction have used vaginal photoplethysmography that measures vaginal pulse amplitude (VPA).
20. **d. Symptoms should cause clinically significant distress.** Regarding the DSM-5 definitions and criteria for sexual dysfunctions in women, all disorders except genitopelvic pain-penetration disorder require that the symptoms meet the DSM-5 definition of that condition, have been present for 6 months on at least 75% of sexual occasions, cause clinically significant distress, are not a consequence of a nonsexual mental disorder or of a severe relationship distress or other significant stressors, and are not attributable to the effect of a medication or illness.
21. **b. Arousal disorder.** Shifren et al. (Prevalence of Female Sexual Problems Associated with Distress and Determinants of Treatment Seeking [PRESIDE] study) reported the overall prevalence of FSD

associated with low desire to be 37.7% in a study involving 50,001 US women and 31,531 respondents aged 18 to 102 years who were evaluated by validated questionnaires. In this study, 8.9% of women aged 18 to 44 years, 12.3% of women aged 45 to 64 years, and 7.4% of women older than 65 years exhibited low desire accompanied by clinically significant distress (HSDD).

22. **c. 5-HT1A receptor agonist and 5-HT 2A receptor antagonist.** Flibanserin is currently the only US Food and Drug Administration (FDA)-approved medication that can be used for the management of premenopausal women with low sexual desire. All of the other medical treatment options that can be applied in patients diagnosed with HSDD are used off-label. Flibanserin (Valeant Pharmaceuticals North America LLC, Bridgewater, NJ), dosed at 100 mg PO once daily at bedtime, is a nonhormonal, centrally acting, postsynaptic 5-HT1A receptor agonist and 5-HT2A receptor antagonist that results in a decrease in serotonin activity and an increase in dopamine and norepinephrine activity.
23. **d. DSM-5 definition of FOD does not contain the criterion that requires orgasmic difficulty to occur despite a normal excitement phase.** FOD can be diagnosed concurrently with at least one other sexual dysfunction. It has been estimated that among women with FOD, 31% also report difficulties with sexual arousal, 18% with lubrication, 14% with desire, 12% with pain, and 0.9% with vaginismus. The DSM-5 definition of FOD does not contain the DSM-IV-TR criterion requiring that difficulty with orgasm occur despite “a normal excitement phase.” This revision was based on the fact that FOD can occur despite intact arousal. A comprehensive sexual history is essential to diagnose and characterize FOD. Nature and onset of the problem should be carefully assessed as well as the other aspects of female sexual function, since FOD often accompanies problems with interest and arousal. Questionnaires alone are not sufficient to diagnose FOD. However, they can be used to track progress of the orgasmic function and monitor treatment response. Systemic testosterone treatment (either alone or in combination with systemic/intravaginal estrogens) and tibolone (a synthetic steroid

available in Europe, which provides a mixture of estrogenic, progestonic, and androgenic actions) have a positive influence on all aspects of female sexual response (desire, arousal, and orgasm) in postmenopausal women with low testosterone/low estrogen levels. However, more research is warranted about the safety and efficacy of hormonal treatment in women with FOD as the primary complaint. SSRIs, which represent the cornerstone of pharmacotherapy of depression, have the potential to adversely affect various aspects of female sexual response cycle, including desire, arousal, and/or orgasm with estimates of 30% to 70% of patients on SSRIs reporting some degree of sexual dysfunction.

24. **b. Symptoms of sexual arousal can be triggered by no stimulus at all.** The International Consultation on Sexual Medicine (ICSM) definition for persistent genital arousal disorder in women is a spontaneous, intrusive, and unwanted genital arousal (i.e., tingling, throbbing, pulsating) in the absence of sexual interest and desire. Any awareness of subjective arousal is typically, but not invariably, unpleasant. The arousal is unrelieved by at least one orgasm, and the feeling of arousal persists for hours or days. In 2001, Leiblum and Nathan defined this syndrome according to the following criteria.
    - Physiologic sexual arousal (genital and breast vasocongestion and sensitivity) persists for an extended period (from hours to days) and does not remit on its own.
    - The signs of physiologic sexual arousal do not remit with ordinary orgasmic experience.
    - The signs of physiologic sexual arousal are experienced in the absence of subjective feelings of sexual desire and arousal.
    - The symptoms of sexual arousal can be triggered by sexual-related stimulus but also by nonsexual cues or no stimulus at all.
    - These symptoms are perceived as intrusive and unwanted, leading at least to some degree of distress.
25. **a. Patients with primary PVD appear to have greater nerve fiber density and thicker vulvar vestibule compared to the secondary group.**

- An important difference between primary and secondary PVD is in the nerve fiber density and the presence of inflammation. Respectively, primary PVD patients appear to have greater nerve fiber density and thicker vulvar vestibule compared to the secondary group. Moreover, hypertonicity of pelvic floor muscles appears to be highly prevalent in patients with PVD. Pelvic muscles dysfunction in patients with PVD can present with decreased muscular strength, reduced speed of contraction, coordination, and reduced endurance.
- Primary PVD: Pain experienced since the first attempt of any type of vaginal penetration (i.e., tampons or sexual activities.) This condition has an early onset, the duration of the pain is longer, and the women are less likely to have had children. They also have a positive family history of the same condition. Frequently there is also an association with enuresis and dysmenorrhea. The presence of local signs of inflammation is less common.
- Secondary PVD: Pain experienced after a certain period of time pertaining to vaginal penetrative activities that previously were pain free. This condition is typically acquired and has a late onset. Larger areas of vestibule are involved in the pain process. The clitoral hood is the area with the most severe pain. Secondary PVD appears to be more frequent than the primary one (65% vs. 35%) and the primary PVD appears to be more frequent in Hispanic women.

Chapter review

1. The vagina is acidic, with a pH between 4 and 5, and is colonized by microorganisms that produce lactic acid.
2. Testosterone production in women comes directly from the ovaries and adrenal glands. Unlike estrogen and progesterone levels, which fall abruptly with menopause, testosterone levels diminish gradually throughout life.

3. Sexual neutrality or being receptive to rather than initiating sexual activity is considered a normal variation of female sexual functioning.
4. Women with incontinence are up to three times more likely to experience decreased arousal, infrequent orgasms, and increased dyspareunia.
5. Lack of estrogen may not directly impair female arousal and desire, but it impairs sexual function by resulting in a decreased vasocongestion and lubrication and increased vaginal epithelial atrophy. Estrogens maintain female genital tissue integrity and thickness.
6. SSRIs have an inhibitory effect on sexual desire, arousal, and orgasm.
7. Optimal female sexual health requires physical, emotional, and mental well-being.
8. Hormonal contraception, prolactin, and elevated sex hormone–binding globulin (SHBG) all tend to decrease bioavailable testosterone.